Physical state spaces and Gaussian observables for quantum observers in de Sitter space

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Abstract

We study two quantum observers and free principal-series matter on two-dimensional de Sitter space in the strict $G_N \to 0$ limit. We specify a weak positive-frequency condition directly on finite gauge-fixed position states and identify the resulting maximal space with the closure of actual group-averaged observer wave packets. The analytic step is a complete position-test core controlled by a fractional radial moment: a low-energy spectral cutoff converges in the orbit norm although a first-moment estimate would not suffice. The identification retains both continuous Clebsch–Gordan channels and all internal discrete paths. With explicit Gaussian proper-time smearing and repeated observer mass projections, bounded equivariant creation blocks descend through a common group cutoff. Their symmetric field is essentially self-adjoint on the full finite-particle domain. On explicit seed tests, the creation block also admits an absolutely convergent position pairing for every physical input; auxiliary position and group cutoffs can be removed in that pairing. For annihilation, finite emitted-mode position integrals converge in physical dual norm at each seed test; on physical seed inputs, a positive Fejér regulator also gives an absolutely convergent uncut position formula. We also record the connection to ordered four-point functions of two independent matter species. These are statements about a specified weak-frequency realization and regulated observables, not a construction of a finite-G_N holographic dual.

1 Introduction and scope

The two-observer model of Kolchmeyer and Liu [1] makes observer recoil and relational correlators accessible while the background remains fixed. An important distinction in this model is between a kinematical Hilbert space, non-normalizable invariant coefficients, and a physical Hilbert space defined by a gauge-fixed position norm. Statements about one of these spaces need not extend to the others by ordinary Hilbert-space density. In particular, evaluation on a continuous spectral diagonal is not a bounded operation on the kinematical tensor product.

We work with de Sitter radius $L = 1$, the proper group $G = SO^+(2,1)$, and real principal-series matter parameters $\nu > 0$. The physical observer mass associated with a representation label s is

$$m(s) = \sqrt{s^2 + \tfrac{1}{4}}. \tag{1.1}$$

Unless a zero-threshold extension is explicitly stated, each actual observer has a fixed lower label $s_* > 0$. This restriction is never imposed on an internal matter or recoupling label. Parity and anti-linear time reversal are not gauged here. The scalar field is free; observer recoil at $G_N \to 0$ is not finite-G_N gravitational scattering.

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Our construction uses the complete unitary principal–principal and principal–discrete decompositions in Appendix E of [1], the group averaging normalization of [2], one-dimensional scattering theory [3, 4], and the analytic-vector criterion [5]. These are established inputs. This draft records the explicit chain of domain estimates and identifications for the chosen realization; no priority claim is made. A reconstructed local vertex, rather than an unannounced alteration of a printed formula, is used where needed; its conventions and source discrepancy are recorded in Appendix B.

The central state-space statement is

$$\mathcal{H}_N^{\max,w}(s_*) = \mathcal{H}_N^{\mathrm{gen}}(s_*) = J_N \mathcal{H}_N^{\mathrm{spec}}(s_*). \tag{1.2}$$

The left-hand side is defined by independent position-test conditions, not by the generated closure. The superscript w is essential: an unspecified formal mode sum is not assigned a convergence topology by this notation. Sections 2–4 prove the identity. Section 5 constructs the regulated weak field. The direct position interpretation of its creation block is proved in Section 6, and the annihilation mode limit in Section 7. The four-point interpretation and its numerical limits are stated in Section 8. Sections 9 and 10 retain the successive reduction steps; their compact-limit hypothesis and assembly question are discharged in Section 11. Section 12 then gives the logarithmic-time packet limit and its ordered early/late plateaux. The full holographic problem is not settled by these strict-$G_N \to 0$ results.

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2 Position norms and a controlled group average

Let $X = \mathrm{dS}_2$ with global coordinates (t, θ) and

$$\mathrm{d}x = \cosh t \; \mathrm{d}t \; \mathrm{d}\theta, \qquad \theta \in [0, 2\pi). \tag{2.1}$$

Let $\mathcal{K}$ be a separable Hilbert space carrying a strongly continuous unitary representation V of G. It will subsequently be a fixed matter-particle sector. On $Y = X \times X$ the diagonal action is $(U(g)\psi)(y) = V(g)\psi(g^{-1}y)$.

Outside the null set where the two embedding vectors have inner product ± 1, the position pairs form six free orbit families. Take the left point to be $(0, 0)$ and the right point to be one of

$$(\pm\tau, 0), \; (\pm\tau, \pi), \quad \tau > 0; \qquad (0, \pm\alpha), \quad 0 < \alpha < \pi. \tag{2.2}$$

Writing this disjoint set of sections as Z and its section map as $s(z)$, the measure decomposition is

$$\mathrm{d}y = \mathrm{d}g \; \mathrm{d}\mu(z), \qquad \mathrm{d}\mu = \sinh\tau \; \mathrm{d}\tau \quad \text{or} \quad \sin\alpha \; \mathrm{d}\alpha. \tag{2.3}$$

The normalization agrees with $\mathrm{d}g = \mathrm{d}\theta_1 \, \mathrm{d}\theta_2 \sinh\lambda \, \mathrm{d}\lambda$. Near the identity the two-boost/rotation Jacobian for the first type of section has absolute value $\tanh\tau$; multiplication by the position density $\cosh\tau$ gives $\sinh\tau$. The spacelike section gives $|\sin\alpha|$. This fixes the constant without dividing by an infinite group volume.

Define the ambient gauge-fixed space $\mathcal{F}(\mathcal{K}) = L^2(Z, \mathrm{d}\mu; \mathcal{K})$. For $\psi \in L^2(Y; \mathcal{K})$ set

$$F_\psi(g, z) = V(g^{-1})\psi(gs(z)), \tag{2.4}$$

$$a_\psi(z) = \int_G \|F_\psi(g, z)\| \, \mathrm{d}g, \tag{2.5}$$

$$\mathcal{D}_{\mathrm{orb}} = \{\psi \in L^2(Y; \mathcal{K}) : a_\psi \in L^2(Z)\}. \tag{2.6}$$

On this domain define $B\psi(z) = \int_G F_\psi(g, z) \, \mathrm{d}g$ as a Bochner integral and its invariant extension by $(\eta\psi)(gs(z)) = V(g)B\psi(z)$.

Proposition 2.1 (Orbit Fubini identity)**.** *For $\phi, \psi \in \mathcal{D}_{\mathrm{orb}}$, the group average is absolutely convergent and*

$$\int_G \langle \phi, U(g)\psi \rangle \, \mathrm{d}g = \langle B\phi, B\psi \rangle_{\mathcal{F}(\mathcal{K})} . \tag{2.7}$$

For inverse-symmetric compact cutoffs $0 \le \chi_R \le 1$ tending to one, the orbit integrals $B_R\psi$ converge to $B\psi$ in $\mathcal{F}(\mathcal{K})$.

Proof. Unimodularity and a Haar change of variables give

$$\int_G |\langle \phi, U(g)\psi\rangle| \, \mathrm{d}g \le \int_Z a_\phi(z) a_\psi(z) \, \mathrm{d}\mu(z) \le \|a_\phi\|_2 \, \|a_\psi\|_2 \, . \tag{2.8}$$

Fubini separates the two Bochner integrals and gives (2.7). The cutoff error is bounded by $\| \int_G (1 - \chi_R(g)) \, \|F_\psi(g, \cdot)\| \, \mathrm{d}g \|_2$, which tends to zero by dominated convergence. A finite-cutoff group average is generally not invariant. An invariant extension of its section values is a different object and is not presumed to retain positive frequency. □

For single angular modes u_L, u_R and $v \in \mathcal{K}$, let $f_i = |u_i|$ and $A_{f_i}(g) = \langle f_i, U_X(g) f_i \rangle$. Nonnegative Tonelli and (2.3) give the exact identity

$$\|a_{u_L u_R v}\|_2^2 = \|v\|^2 \int_G A_{f_L}(g) A_{f_R}(g) \, \mathrm{d}g. \tag{2.9}$$

No decay assumption on a matter matrix coefficient is used.

Lemma 2.2 (Finite position states are distributions)**.** *For every $k \in \mathcal{F}(\mathcal{K})$, the extension $Rk(gs(z)) = V(g)k(z)$ lies in $L^1_{\mathrm{loc}}(Y; \mathcal{K})$. The map R is continuous and injective. For every $\phi \in \mathcal{D}_{\mathrm{orb}}$,*

$$T_k(\phi) := \int_Y \langle Rk(y), \phi(y) \rangle \, \mathrm{d}y = \langle k, B\phi \rangle \, , \qquad |T_k(\phi)| \le \|k\|_2 \, \|a_\phi\|_2 \, . \tag{2.10}$$

Proof. The radial bound below implies that a compact smooth position test belongs to $\mathcal{D}_{\mathrm{orb}}$, with $\|a_\phi\|_2 \le C_C \sup_C \|\phi\|$ on a fixed compact set C: dominate its norm by two compact radial envelopes and apply (2.9). Tonelli gives

$$\int_Y \|Rk(y)\| \, \|\phi(y)\| \, \mathrm{d}y = \int_Z \|k(z)\| \, a_\phi(z) \, \mathrm{d}\mu(z). \tag{2.11}$$

A compact cutoff equal to one on C proves the local L^1 bound and (2.10). If $Rk = 0$ almost everywhere, integrate over the image of a finite positive Haar-measure set times Z in (2.3); then $k = 0$ almost everywhere. No trace of an arbitrary L^1 function on a section is taken. □

For a smooth noncompact orbit test, (2.10) is also the limit of compact smooth position cutoffs. The orbit norm of $(1 - \xi_j)\phi$ tends to zero when $0 \le \xi_j \le 1$ tend to one. Thus these pairings are determined by the position distribution.

3 A complete spectral core in the orbit norm

For a single angular mode use the unitary radial variable

$$u(t, \theta) = \frac{e^{in\theta}}{\sqrt{2\pi \cosh t}} Q(t), \qquad b_\alpha(u) = \|Q\|_2 + \int_{\mathbb{R}} (1 + |t|)^\alpha |Q(t)| \, \mathrm{d}t. \tag{3.1}$$

Write the last integral as $M_\alpha(Q)$. The Casimir minus 1/4 becomes

$$H_k = -\partial_t^2 + (1/4 - k^2) \operatorname{sech}^2 t, \qquad k = |n|, \quad D(H_k) = H^2(\mathbb{R}). \tag{3.2}$$

All scattering estimates below are at fixed k unless stated otherwise.

3.1 The radial estimate for absolute orbit integration

The repulsive operator H_0 has no bound states and no zero resonance. For the latter, a bounded solution of $H_0 h = 0$ has $h'(\pm\infty) = 0$ and $h' = O(e^{-2|t|})$ by integration of the equation. Integration by parts then gives $\int (|h'|^2 + \frac{1}{4}\operatorname{sech}^2 t |h|^2)\,\mathrm{d}t = 0$, so $h = 0$.

Choose Jost solutions $F_{0,+} \sim e^{ist}$ at $+\infty$ and $F_{0,-}(t,s) = F_{0,+}(-t,s)$:

$$F_{0,+}(t,s) = e^{ist}\, {}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2}; 1 - is; \tfrac{1-\tanh t}{2}\right), \tag{3.3}$$

$$T_0(s) = \frac{\Gamma(1/2 - is)^2}{\Gamma(1-is)\Gamma(-is)}, \qquad |T_0(s)|^2 = \tanh^2 \pi s, \tag{3.4}$$

$$e_{0,\pm}(t,s) = (2\pi)^{-1/2} T_0(s) F_{0,\pm}(t,s). \tag{3.5}$$

The distorted Fourier transform is unitary with two channels and measure $\mathrm{d}s$, by [3, Theorem 2.3]. Volterra estimates on the good half-axis and the scattering relation on the other half-axis give

$$|e_{0,\pm}(t,s)| \le C \min\{1, s(1+|t|)\}. \tag{3.6}$$

For example $|F_{0,+}| \le C$ on $t \ge 0$ and $|F_{0,+}| \le C(1+|t|)$ at low energy on $t \le 0$; $T_0(s) = O(s)$ yields the second bound. The identity $T_0 F_{0,+} = \overline{F_{0,-}} + R_0 F_{0,-}$, with $|R_0| \le 1$, yields the uniform bound on the other half-axis.

Let $f = |u|$, which is rotation invariant. Its radial transform is $|Q|$, and let q_f be the sum of the squares of its two spectral coefficients. For $0 < \alpha \le 1$, (3.6) gives

$$q_f(s) \le C M_\alpha(Q)^2 \min(s^{2\alpha}, 1), \qquad \int_0^\infty q_f(s)\,\mathrm{d}s = \|Q\|_2^2 . \tag{3.7}$$

The spherical subrepresentation of $L^2(X)$ has principal spectrum of multiplicity two: the radial Casimir is purely absolutely continuous above $1/4$, while a final discrete representation has no $K = SO(2)$ fixed vector. Its unit spherical matrix coefficient is $P_{-1/2+is}(\cosh\lambda)$, with group Schur weight [2, 6]

$$K_{\mathrm{P}}(s) = \frac{4\pi^2}{s \tanh \pi s} = \rho_{\mathrm{Pl}}(s)^{-1}. \tag{3.8}$$

The radial and group multiplicity bases differ by a measurable unitary matrix, leaving q_f unchanged. Consequently

$$\|A_f\|^2_{L^2(G)} = \int_0^\infty K_{\mathrm{P}}(s) q_f(s)^2\,\mathrm{d}s \le C_\alpha b_\alpha(u)^4, \qquad \tfrac{1}{4} < \alpha \le 1. \tag{3.9}$$

At zero the integrable power is $s^{4\alpha-2}$; above one use the boundedness of K_{P}, the bound on q_f, and Parseval. The equality is first applied to smooth compact spectral densities and then extended by simultaneous $L^1(\mathrm{d}s)$ and $L^2(K_{\mathrm{P}}\,\mathrm{d}s)$ approximation. This does not exchange arbitrary distributional Schur kernels.

Combining (2.9) and (3.9) yields

$$\|a_{u_L u_R v}\|_2 \le C_\alpha b_\alpha(u_L) b_\alpha(u_R)\, \|v\| . \tag{3.10}$$

This constant is independent of the angular labels, since it uses the radial operator of the absolute values. It proves the compact-test estimate used in Lemma 2.2.

3.2 All angular sectors, including bound states

Set $\ell_k = k - 1/2$ and $A_k = \partial_t + \ell_k \tanh t$. Then

$$A_k^* A_k = H_k + \ell_k^2, \qquad A_k A_k^* = H_{k-1} + \ell_k^2. \tag{3.11}$$

The second operator is strictly positive, with lower bound $1/4$ for $k=1$ and $2k-2$ for $k\geq 2$. Polar decomposition gives an isometry onto $(\ker A_k)^\perp$,

$$V_k = A_k^*(H_{k-1}+\ell_k^2)^{-1/2}, \qquad \ker A_k = \operatorname{span}\{\operatorname{sech}^{\ell_k} t\}. \tag{3.12}$$

It intertwines H_{k-1} and H_k. Induction proves the full decomposition

$$H_k \simeq H_0 \oplus \bigoplus_{\Delta=1}^{k} \{-(\Delta-1/2)^2\}. \tag{3.13}$$

The bound states $b_{k,\Delta}$ are exponentially decreasing and follow from the same intertwiners. Their Casimir is $\Delta(1-\Delta)$. The no-bound-state hypothesis of [3] was used only for H_0.

The Jost solutions and scattering coefficients are

$$F_{k,+}(t,s) = e^{ist}\,{}_2F_1\left(\tfrac{1}{2}-k, \tfrac{1}{2}+k; 1-is; \tfrac{1-\tanh t}{2}\right), \tag{3.14}$$

$$T_k(s) = T_0(s)(-1)^k \prod_{j=1}^{k} \frac{j-1/2-is}{j-1/2+is}, \tag{3.15}$$

$$R_k(s) = -\frac{i(-1)^k}{\sinh \pi s} T_k(s). \tag{3.16}$$

Thus $T_k(s) = -i(-1)^k \pi s + O(s^2)$ and $R_k(0) = -1$. With $W_k = F_{k,-}F'_{k,+} - F'_{k,-}F_{k,+}$, $W_k(0) = 2(-1)^{k+1}/\pi \neq 0$, excluding a zero resonance. The normalized modes $e_{k,\pm} = (2\pi)^{-1/2} T_k F_{k,\pm}$ are the unitary continuous basis: applying V_k to the preceding modes multiplies them by the unit phases $\mp(\ell_k - is)/\sqrt{\ell_k^2+s^2}$.

3.3 Spectral cutoff in a stronger position topology

For $Q \in C_c^\infty(\mathbb{R})$ its continuous coefficients satisfy

$$\begin{aligned} a_\pm(s) = \int \overline{e_{k,\pm}(t,s)} Q(t)\,\mathrm{d}t = s\beta_\pm(s) \quad \text{near } 0, \qquad \beta_\pm \in C^\infty, \\ \sup_{s\geq 1} s^L |\partial_s^j a_\pm(s)| < \infty \quad \text{for every } j, L. \end{aligned} \tag{3.17}$$

The zero follows from $T_k = O(s)$ on compact position sets. For high energy, integrate H_k^M onto Q and use compact-position bounds on the spectral derivatives of $e_{k,\pm}$.

These derivative bounds follow from a uniform Volterra estimate. On $t \geq 0$ write $F_{k,+} = e^{ist} m_+$; the kernel for m_+ is $D_s(v) = \int_0^v e^{2isr}\,\mathrm{d}r$. Since $|\partial_s^j D_s(v)| \leq 2^j v^{j+1}/(j+1)$ and the potential has all polynomial moments with an exponential tail, successive parameter differentiation yields

$$\sup_{s\geq 0} |\partial_s^j (m_+(t,s)-1)| \leq C_{k,j} e^{-2t}. \tag{3.18}$$

The other good half-axis is identical. Gamma-ratio asymptotics in (3.16) give bounded derivatives of T_k, R_k on $[0,\infty)$. On the bad half-axis the scattering relation implies that derivatives of the full modes grow only polynomially in $|t|$.

Choose $\chi = 1$ on $[0,1]$, $\chi = 0$ on $[2,\infty)$, smooth in between. The low-spectral part is

$$L_\epsilon Q(t) = \sum_{\pm} \int_0^\infty e_{k,\pm}(t,s)\chi(s/\epsilon) a_\pm(s)\,\mathrm{d}s. \tag{3.19}$$

On each half-axis its plane-wave amplitudes have the form $c_\epsilon(s) = s\chi(s/\epsilon)B(s)$ with the first two derivatives of B bounded. Hence

$$c_\epsilon(0) = 0, \quad \|c_\epsilon\|_1 = O(\epsilon^2), \quad |c'_\epsilon(0)| + \|c''_\epsilon\|_1 = O(1). \tag{3.20}$$

Two integrations by parts retain the second endpoint term and give

$$|L_\epsilon Q(t)| \le C\min(\epsilon^2, |t|^{-2}) + C\epsilon^2 e^{-2|t|} \le \frac{C'\epsilon^2}{(1+\epsilon|t|)^2}. \tag{3.21}$$

For $0 \le \alpha < 1$,

$$M_\alpha(L_\epsilon Q) = O(\epsilon^{1-\alpha}), \qquad \|L_\epsilon Q\|_2 = O(\epsilon^{3/2}). \tag{3.22}$$

This controls the linked large-position/low-energy region.

At high energy (3.17) permits arbitrarily many integrations by parts without endpoint terms. The tail beyond a smooth cutoff at R is bounded by $C_{L,M}R^{-L}(1+|t|)^{-M} + C_L R^{-L}e^{-2|t|}$. Retaining all bound states, define

$$\begin{aligned} Q_{\epsilon,R}(t) = &\sum_{\Delta=1}^{k} \langle b_{k,\Delta}, Q\rangle\, b_{k,\Delta}(t) \\ &+ \sum_{\pm} \int_0^\infty e_{k,\pm}(t,s)(1-\chi(s/\epsilon))\chi(s/R)a_\pm(s)\,\mathrm{d}s. \end{aligned} \tag{3.23}$$

For $R > 2\epsilon$ its error is exactly the low plus high tail. Thus

$$\|Q - Q_{\epsilon,R}\|_2 + M_\alpha(Q - Q_{\epsilon,R}) \longrightarrow 0, \quad \epsilon \downarrow 0,\ R \uparrow \infty, \quad \alpha < 1. \tag{3.24}$$

We henceforth take $1/4 < \alpha < 1$, for example $\alpha = 1/2$.

Theorem 3.1 (Completeness of the orbit-test core)**.** *Let $\mathcal{S}_{\mathrm{all}}$ be the span of products of two fixed-angular-mode compact smooth continuous-spectral packets or bound modes, with arbitrary $v \in \mathcal{K}$. It is contained in $\mathcal{D}_{\mathrm{orb}}$. Every $\phi \in C_c^\infty(Y;\mathcal{K})$ has approximants $\psi_j \in \mathcal{S}_{\mathrm{all}}$ such that*

$$\|\phi - \psi_j\|_{L^2(Y;\mathcal{K})} + \left\|a_{\phi-\psi_j}\right\|_2 \longrightarrow 0. \tag{3.25}$$

In particular, $\overline{B\mathcal{S}_{\mathrm{all}}} = \mathcal{F}(\mathcal{K})$.

Proof. On a common compact position set, approximate ϕ uniformly by finite matter-rank, finite angular-Fourier, smooth radial tensor sums. Fixed radial cutoffs retain compact support. Lemma 2.2 controls this orbit error. For each of the finitely many factors use (3.24) and (3.10). A diagonal choice gives the limit. If k is orthogonal to $B\mathcal{S}_{\mathrm{all}}$, it is orthogonal to all $B\phi$ with ϕ compact smooth. Equation (2.10) gives $Rk = 0$ as a distribution; injectivity gives $k = 0$. □

Remark 3.2 (A hard interior cutoff is different). Deleting a narrow band around $s_* > 0$ need not converge in this topology. An amplitude nonzero at s_* gives a packet of scale $\epsilon f(\epsilon t)e^{is_* t}$, whose M_α generally scales as $\epsilon^{-\alpha}$. The zero in (3.17) was essential. The actual threshold will be imposed after identifying the physical spectral Hilbert space.

4 Weak frequency constraints and the physical space

4.1 The single-observer frequency split

Let Ψ_n^s be the normalized positive-frequency modes of [1], and $\widetilde{\Psi}_n^s(t,\theta) = (-1)^n\Psi_n^s(-t,\theta)$. Appendix A proves their wave-packet normalization and mutual orthogonality by radial unitary evolution. They span the full continuous multiplicity-two spectrum at each angular label. Together with the bound modes of (3.13), they form the core of Theorem 3.1.

Let $H_+ \subset L^2(X)$ be the original positive-frequency subspace and $H_\perp = H_+^\perp$. Both are G-invariant. Write $\mathcal{S}_+$ for the finite span of positive-frequency packets with coefficients in

$C_c^\infty((0,\infty))$ and $\mathcal{S}_\perp$ for the negative-frequency packets and bound modes. For $\mathcal{K}_N = \mathrm{Sym}^N P_\nu$ define

$$\mathcal{E}_{\epsilon\delta} = \mathrm{span}\{u_L u_R v : u_L \in \mathcal{S}_\epsilon,\ u_R \in \mathcal{S}_\delta,\ v \in \mathcal{K}_N\}, \quad \epsilon,\delta \in \{+,\perp\}. \tag{4.1}$$

Their B images are mutually orthogonal: between distinct sectors one observer factor in each group matrix coefficient is zero. Absolute group averaging and Theorem 3.1 give

$$\mathcal{F}(\mathcal{K}_N) = \mathcal{F}_{++} \oplus \mathcal{F}_{+\perp} \oplus \mathcal{F}_{\perp+} \oplus \mathcal{F}_{\perp\perp}, \quad \mathcal{F}_{\epsilon\delta} = \overline{B\mathcal{E}_{\epsilon\delta}}. \tag{4.2}$$

This constructs a bounded physical frequency projection. It does not evaluate a kinematical projection at a measure-zero trivial representation in a regular Plancherel decomposition.

Definition 4.1 (Weak positive frequency)**.** A state $k \in \mathcal{F}(\mathcal{K}_N)$ is weakly positive frequency in both observers if

$$T_k(\phi) = 0 \quad \text{for every } \phi \in \mathcal{E}_{+\perp} + \mathcal{E}_{\perp+} + \mathcal{E}_{\perp\perp}. \tag{4.3}$$

The pairings are the absolute position integrals (2.10).

This definition uses only single-observer frequency and position tests. By (4.2), its solution space is exactly $\mathcal{F}_{++}$.

4.2 Actual product amplitudes and all multiplicities

For $N \geq 1$, complete normalized CG analysis gives at fixed actual right-observer label a

$$(P_a \otimes \mathrm{Sym}^N P_\nu)^{\mathrm{P}} \simeq \int_{d>0}^{\oplus} \mathrm{d}d\, (P_d \otimes \mathcal{M}_{N,d,a}). \tag{4.4}$$

Inductively add one principal-series particle at a time. Only principal–principal and principal–discrete decompositions are needed. The final discrete summands are retained separately; internal discrete paths remain in $\mathcal{M}$. Bose symmetrization is a decomposable orthogonal projection on multiplicity, creating no new spectral type. The original magnetic basis trivializes the measurable field in a; explicit CG coefficients and their removable limits give jointly measurable transforms.

Each P_d has one unit $K = 0$ vector, while a final discrete representation has none. Hence

$$U_{N,a} : (P_a \otimes \mathrm{Sym}^N P_\nu)^{(0)} \longrightarrow \int_{d>0}^{\oplus} \mathrm{d}d\, \mathcal{M}_{N,d,a} \tag{4.5}$$

is unitary. Take a countable original finite-mode Bose basis $e_i(a)$ and write $u_i(a,d) = U_{N,a} e_i(a)$, so $\int \|u_i(a,d)\|^2 \,\mathrm{d}d = 1$.

For a compact smooth left packet f of magnetic label zero and an arbitrary right total-$K = 0$ vector v, the group-averaged amplitude obeys

$$g_{f,v}(d) = K_{\mathrm{P}}(d) f(d) v(d), \qquad \|g_{f,v}\|_{\mathrm{spec}} \leq B_f \|v\|, \qquad B_f^2 = \operatorname*{ess\,sup}_{d>0} K_{\mathrm{P}}(d) |f(d)|^2. \tag{4.6}$$

The left mass integration must be performed first. Indeed

$$|D^s_{00}(g)| \leq \Xi_0(\lambda) := P_{-1/2}(\cosh\lambda) \leq C(1+\lambda) e^{-\lambda/2} \tag{4.7}$$

uniformly in $s > 0$, by the circle integral of the Legendre function. Compact left spectral integration gives the left coefficient $L(g)$ decay $e^{-\lambda/2}(1+\lambda)^{-r}$ up to exponentially smaller terms. For $r \geq 3$, $\int_G |L| \Xi_0 \,\mathrm{d}g < \infty$. The right cross-spectral density is in $L^1(\mathrm{d}d)$ by Cauchy–Schwarz, so Fubini is absolute. The wave-packet Schur relation gives (4.6); no absolute integration at two fixed masses is asserted.

For the actual original product $v(a) = k_R(a)e_i(a)$,

$$g_{i;f,k_R}(d,a) = K_{\mathrm{P}}(d)f(d)k_R(a)u_i(a,d). \tag{4.8}$$

These amplitudes are dense in

$$\mathcal{H}_N^{\mathrm{spec},0} = \int_{d>0}^{\oplus} \rho_{\mathrm{Pl}}(d)\,\mathrm{d}d \int_{a>0}^{\oplus} \mathrm{d}a\,\mathcal{M}_{N,d,a}. \tag{4.9}$$

To prove this, take h orthogonal to all (4.8). Since $\rho_{\mathrm{Pl}}K_{\mathrm{P}} = 1$, we have

$$\int f(d)k_R(a)\,\langle h(d,a), u_i(a,d)\rangle\,\mathrm{d}a\,\mathrm{d}d = 0.$$

On compact $A \times D$ the absolute integral is bounded by $\|h\|_{\mathrm{spec}}\sqrt{|A|\sup_D \rho_{\mathrm{Pl}}^{-1}}$. Product-test totality implies that the pairing vanishes almost everywhere for all countably many i simultaneously. Restrict h to $A \times D$ and apply $U_{N,a}^{-1}$; every original basis coefficient vanishes. Compact exhaustion gives $h = 0$, without a bounded diagonal-trace map.

For all finite left magnetic labels the amplitude is

$$g_\psi(d) = K_{\mathrm{P}}(d)\sum_j (-1)^{m_j} f_j(d)v^{\mathrm{P}}_{j,-m_j}(d). \tag{4.10}$$

These terms add coherently. In the compact picture every unit magnetic-basis matrix coefficient is bounded by Ξ_0, so the same left-packet estimate applies. At fixed right total magnetic label only finitely many final discrete representations occur; exponential decay makes their mixed integrals absolute, and cross-series Schur terms vanish. Both norms and cross-pairings of the full product class therefore equal the spectral inner product. The dense small class (4.8) has the same FP closure as $B\mathcal{E}_{++}$.

For $N = 0$ the right total label is $d = a$. The spectral space is $L^2(\rho_{\mathrm{Pl}}(d)\,\mathrm{d}d)$ and $g_{f,k_R} = K_{\mathrm{P}}fk_R$ has dense image. There is no independent $\mathrm{d}a$ integral. We obtain a concrete surjective isometry $J_N^0 : \mathcal{H}_N^{\mathrm{spec},0} \to \mathcal{F}_{++}$ for every finite N, specified on actual seeds by $J_N^0 g_\psi = B\psi$.

4.3 Observer thresholds and maximality

Let $\mathcal{S}_{+,<}$ contain positive-frequency packets supported in $(0, s_*)$. In addition to (4.3), require

$$T_k(u_L u_R v) = 0 \tag{4.11}$$

when either $u_L \in \mathcal{S}_{+,<}$ and $u_R \in \mathcal{S}_+$ or conversely. Define $\mathcal{H}_N^{\max,w}(s_*)$ to contain all finite FP states satisfying these two independent test conditions.

Theorem 4.2 (Maximal weak-frequency realization)**.** *Identity* (1.2) *holds as an equality of concrete subspaces of* $\mathcal{F}(\mathcal{K}_N)$*. For* $N \geq 1$*, its spectral space is* (4.9) *restricted to* $a, d \geq s_*$*. For* $N = 0$ *only* $d \geq s_*$ *remains.*

Proof. Equation (4.2) first puts the state in $\mathcal{F}_{++} = J_N^0\mathcal{H}_N^{\mathrm{spec},0}$. Use (4.8) with f supported below s_* and repeat the local L^1/inverse-CG argument to obtain $g = 0$ on $d < s_*$. Taking instead k_R below s_* gives $g = 0$ on $a < s_*$. Conversely, (4.10) retains left support as d support and right support as a support, since CG analysis is at fixed a. Every low-observer-spectrum test is therefore orthogonal to a high-supported g. The fixed-threshold actual product construction has dense image in exactly that spectral subspace. The vacuum follows from $K_{\mathrm{P}}fk_R$. Threshold atoms are not additional vectors in this identified L^2 space. □

The topology qualification is substantive. Equation (4.40) of [1] does not prescribe a test topology for an arbitrary non-normalizable coefficient sum. If it admits the absolute test pairings here with the stated positive-frequency support, Theorem 4.2 applies. It is not a theorem about every formal summation prescription.

There is also a sufficient limit criterion. Suppose $k_j \in \mathcal{H}_N^{\mathrm{gen}}(s_*)$, $\sup_j \|k_j\|_{\mathrm{FP}} < \infty$, and $Rk_j \to F$ locally as distributions. Hilbert weak compactness gives a weakly convergent subsequence. Lemma 2.2 identifies its limit as $F = Rk$. The generated space is weakly closed, so k belongs to it; injectivity makes every weak cluster point the same. The full sequence converges weakly and $\|k\| \le \liminf_j \|k_j\|$. Convergence of the norms to $\|k\|$ gives strong convergence. A local distributional limit without a uniform FP bound is not covered.

5 Gaussian local kernels and their weak descent

We now keep $s_* > 0$, a real matter label $\nu > 0$, and a proper-time width $\sigma > 0$ fixed. Start with the original finite-mode local matrix element of [1, (5.4)], not an assumed unregulated physical field:

$$t^{bn_b}_{an_a;\nu n} = \int_X \overline{\Psi^a_{n_a}(x)}\,\overline{\psi^\nu_n(x)}\,\Psi^b_{n_b}(x)\,\mathrm{d}x. \tag{5.1}$$

Here b is the observer label before creation and a after creation. The fixed-mode integral is absolutely convergent on positive compact spectral sets. With the conventions of Appendix B,

$$t^{bn_b}_{an_a;\nu n} = \sqrt{\frac{\pi}{r(\nu)}}\,\overline{\mathcal{V}^b_{\nu,a,\mathrm{rec}}}\,C^{bn_b,S}_{an_a|\nu n}, \qquad r(s) = s\tanh \pi s. \tag{5.2}$$

The coefficient C is a CG synthesis coefficient; its analysis adjoint uses the conjugate. Interchanging them would reverse the local kernel.

5.1 A bounded kinematical creation map

The negative-sign time convention of [1] and a normalized real Gaussian centered at τ give

$$h_{\sigma,\tau}(a,b) = \exp\left[-\frac{\sigma^2}{2}(m(a)-m(b))^2 - i(m(a)-m(b))\tau\right]. \tag{5.3}$$

Set $w(b) = \pi r(b)/4 = \pi^3 \rho_{\mathrm{Pl}}(b)$. At fixed a the normalized S-channel synthesis J_a is an isometry from $\int_{b>0}^{\oplus} P_b\,\mathrm{d}b$ into $P_a \otimes P_\nu$. Its coefficients are $\sqrt{w(b)}C^{bn_b,S}_{an_a|\nu n}$. Define

$$F_{\nu,\sigma,\tau}(a,b) = \sqrt{\frac{\pi}{r(\nu)}}\,\frac{\overline{\mathcal{V}^b_{\nu,a,\mathrm{rec}}}}{\sqrt{w(b)}}\,h_{\sigma,\tau}(a,b), \tag{5.4}$$

$$(T_{\nu,\sigma,\tau}u)(a) = J_a[F_{\nu,\sigma,\tau}(a,\cdot)u(\cdot)], \qquad a,b \ge s_*. \tag{5.5}$$

The full CG transform has other channels; J_a selects just the one local creation vertex. The scalar modulus is

$$|F_{\nu,\sigma,\tau}(a,b)|^2 = V_\nu(a|b)e^{-\sigma^2[m(a)-m(b)]^2}, \tag{5.6}$$

$$V_\nu(a|b) = \frac{a\tanh(\pi a)\cosh^2(\pi b)}{16\pi^5}\prod_{\epsilon_1,\epsilon_2,\epsilon_3=\pm1}\Gamma\left(\frac{1}{4}+\frac{i}{2}(\epsilon_1\nu+\epsilon_2 a+\epsilon_3 b)\right). \tag{5.7}$$

The gamma factors form four conjugate pairs, so the weight is positive. Synthesis isometry and nonnegative Tonelli imply

$$T^*T = A_\sigma(\hat{s}_R), \tag{5.8}$$

$$A_\sigma(b) = \int_{s_*}^{\infty} V_\nu(a|b) e^{-\sigma^2[m(a)-m(b)]^2}\,\mathrm{d}a,$$

$$\|T\|^2 = \operatorname*{ess\,sup}_{b\geq s_*} A_\sigma(b) =: \mathfrak{M} < \infty. \tag{5.9}$$

One sufficient uniform estimate is

$$V_\nu(a|b) \leq C_\nu \frac{e^{-2\pi(a-b)_+}}{1+|a-b|}. \tag{5.10}$$

To obtain it, apply the global bound $|\Gamma(1/4+ix)|^2 \leq C(1+|x|)^{-1/2}e^{-\pi|x|}$ to the four conjugate pairs. The sum of absolute gamma arguments gives exponential suppression at least $e^{-2\pi(a-b)_+}$ after the $\cosh^2(\pi b)$ factor. The two large sum arguments supply the factor $(1+a+b)^{-1}$ that controls the prefactor a; the two difference arguments give $(1+|a-b|)^{-1}$, with constants depending on fixed ν. Bounded regions are absorbed into the constant. On $|a-b| \geq 1$, $|m(a)-m(b)| \geq |a-b|/2$; splitting there proves the uniform integral bound in (5.9).

The map T is a bounded G intertwiner. For $W_N = H_{R,\geq s_*} \otimes \mathrm{Sym}^N P_\nu$ set

$$C_N = \sqrt{N+1}\,(1_R \otimes P_{\mathrm{Sym}^{N+1}})(T\otimes 1), \qquad \|C_N\| \leq \sqrt{(N+1)\mathfrak{M}}. \tag{5.11}$$

This gives the standard normalized Fock creation factor. There is no general identity $C_N^*C_N = (N+1)A_\sigma(\hat{s}_R)$ after symmetrization. In particular $\hat{s}_R$ is not the total representation label d.

5.2 A common group cutoff defines the physical weak kernel

The complete right-system decomposition makes C_N decomposable: $C_N = \int^{\oplus}(1_{P_d}\otimes c_N(d))\,\mathrm{d}d$ on the principal part, with $\|c_N(d)\| \leq \sqrt{(N+1)\mathfrak{M}}$ almost everywhere. Let

$$\mathcal{E}_N^0 = \mathrm{span}\{|f,0\rangle_L \otimes v : f \in C_c^\infty((s_*,\infty)),\ v \in W_N^{(0)}\}. \tag{5.12}$$

The right vector may have infinite magnetic support and noncompact total spectrum. The left-packet argument leading to (4.6) still gives absolute scalar group averaging on this space, with $\mathcal{A}_N(f\otimes v) = K_{\mathrm{P}} f v$ and $Q_{\mathrm{GA}}(S_1,S_2) = \langle \mathcal{A}_N S_1, \mathcal{A}_N S_2\rangle$. Both C_N and C_N^* preserve the corresponding total-$K=0$ test classes. They need not preserve $\mathcal{D}_{\mathrm{orb}}$.

Take $\eta_{N,R} = \int_G \chi_R(g)U_N(g)\,\mathrm{d}g$, with the inverse-symmetric compact cutoffs used previously. The local weak creation form is

$$\mathcal{B}_N^{\mathrm{loc}}(\Phi,\Psi) := \lim_{R\to\infty} \langle \Phi, (1_L\otimes C_N)\eta_{N,R}\Psi\rangle_{\mathrm{kin}}\,. \tag{5.13}$$

Its origin in the local mode integral can be retained explicitly. Choose finite-rank orthogonal projections P_j, Q_j from finite original magnetic modes and compact mass packets in the input and output spaces. Then $Q_jC_NP_j \to C_N$ strongly and uniformly boundedly. Each finite matrix element comes from (5.1) with (5.3). At fixed R take $j\to\infty$ first, and only then take $R\to\infty$. A finite-rank approximation is not assumed to intertwine G.

For the restored bounded intertwiner, finite-R Bochner integration gives $(1\otimes C_N)\eta_{N,R} = \eta_{N+1,R}(1\otimes C_N)$. Absolute scalar group averaging now proves existence and cutoff independence of (5.13). Computing it yields

$$\mathcal{B}_N^{\mathrm{loc}}(\Phi,\Psi) = \int_{d\geq s_*} \rho_{\mathrm{Pl}}(d)\,\langle \mathcal{A}_{N+1}\Phi(d), c_N(d)\mathcal{A}_N\Psi(d)\rangle\,\mathrm{d}d. \tag{5.14}$$

This is an identification of the previously defined form, not its definition by a spectral right-hand side.

The bound on c_N makes the form zero on either physical null space and bounded in the two physical norms. Riesz representation and the isometries J_N therefore produce a unique bounded physical creation block

$$\mathsf{C}_N = J_{N+1} c_N J_N^{-1}, \qquad \|\mathsf{C}_N\| \le \sqrt{(N+1)\mathfrak{M}}. \tag{5.15}$$

The same cutoff applied to C_N^* gives its physical adjoint. An output can also be constructed in position norm: for fixed f, approximate $C_N v$ in the right Hilbert norm by actual original product packets w_j. Equation (4.6) bounds the physical error by $B_f \|w_j - C_N v\|$. Their true orbit averages converge to the Riesz output independently of the approximation.

Theorem 5.1 (Essential self-adjointness of the regulated weak field)**.** *On $\mathcal{H} = \bigoplus_{N\ge 0} \mathcal{H}_N^{\mathrm{gen}}(s_*)$, let $\mathcal{D}_{\mathrm{fin}}$ contain every vector with finitely many nonzero particle sectors, with arbitrary Hilbert vectors in each such sector. The symmetric field $\Phi_{\sigma,\tau} = \mathsf{C} + \mathsf{C}^*$ is essentially self-adjoint on this dense invariant domain.*

Proof. For ψ supported in $N \le r$, the block bound gives

$$\left\|\Phi_{\sigma,\tau}^j \psi\right\| \le (2\sqrt{\mathfrak{M}})^j \sqrt{\frac{(r+j)!}{r!}}\, \|\psi\|\,. \tag{5.16}$$

The analytic-vector series converges for every positive radius. Nelson's criterion [5] proves essential self-adjointness. □

By Theorem 4.2, this is also the field on the specified maximal weak-frequency space. The theorem concerns the common-cutoff weak realization; it does not identify unrestricted position multiplication.

6 A direct position integral for the creation block

We now remove the auxiliary spacetime and group cutoffs from a specific position pairing. The proper-time Gaussian and the actual observer threshold remain fixed. The output is tested by an original seed, not by a second invariant wavefunction integrated over the full gauge orbit. This distinction avoids the group-volume divergence noted after (5.8) of [1]. Transposition to test vectors is standard; the domain estimates below supply the additional input needed here. In particular, the invariant-domain requirements in refined algebraic quantization [7, Section II] cannot be replaced by formal commutation with the constraints. The distinction between operators on invariants and co-invariants is also discussed in [8, Section 5.1].

6.1 Test vectors and observer time

Let $\mathcal{P}_N$ be the finite span of actual product seeds

$$\phi(y) = u_{0,f_L}(x_L)\, u_{n_R,f_R}(x_R)\, v_N, \qquad f_L, f_R \in C_c^\infty((s_*,\infty)), \tag{6.1}$$

where v_N is a finite linear combination of normalized matter occupation vectors and $n_R + J_{\mathrm{m}} = 0$ term by term. These are the small total-$K = 0$ seeds used in Section 4; $B\mathcal{P}_N$ is dense in the physical sector. The case $N = 0$ has $n_R = 0$.

Write $E_R(u) = e^{iuH_R}$ on the kinematical observer space. It induces a strongly continuous physical unitary $\mathsf{U}_N(u)$: in the identified spectrum it multiplies by $e^{ium(a)}$, where a is the actual right-observer label, with $a = d$ for $N = 0$. Thus, on these seeds,

$$\mathsf{U}_N(u) B\psi = B E_R(u)\psi, \qquad \|\mathsf{U}_N(u)k\|_{\mathrm{FP}} = \|k\|_{\mathrm{FP}}\,. \tag{6.2}$$

This construction uses the already identified state space and does not assume a local field. Since the mass phase preserves compact spectral support, $E_R(u)\mathcal{P}_N \subset \mathcal{P}_N$ for every finite u.

For a finite-mode matter vector define the ordinary, finite sum

$$\mathfrak{a}_\nu(x)v = \sum_n \psi_n^\nu(x) a_n v. \tag{6.3}$$

The sum is finite on this domain. Fixed-n mode asymptotics give $\sup_x |\psi_n^\nu(x)| < \infty$; more precisely $|\psi_n^\nu(t,\theta)| \le c_{n,\nu}(\cosh t)^{-1/2}$. Consequently

$$\|\mathfrak{a}_\nu(x)v\| \le \sum_{n\in F_v} c_{n,\nu}\,\|a_n v\| =: c_v. \tag{6.4}$$

No uniform estimate as the occupied mode set F_v grows is asserted.

For $\phi \in \mathcal{P}_{N+1}$ set

$$\zeta_{\phi,u}(y) = \mathfrak{a}_\nu(x_R)\,[E_R(u)\phi](y) \in \mathcal{K}_N. \tag{6.5}$$

Although $\zeta_{\phi,u}$ need not obey the observer positive-frequency condition, it is an ambient orbit test. For a single product in (6.1), its pointwise norm is at most $c_v|u_{0,f_L}(x_L)u_{n_R,e^{ium}f_R}(x_R)|$. The norm inequality survives untwisting because the matter representation is unitary.

Lemma 6.1 (Polynomial orbit bound for the transposed test)**.** *For every fixed $\phi \in \mathcal{P}_{N+1}$ there is a constant C_ϕ such that*

$$\zeta_{\phi,u} \in \mathcal{D}_{\mathrm{orb}}, \qquad \left\|a_{\zeta_{\phi,u}}\right\|_2 \le C_\phi(1+|u|)^2. \tag{6.6}$$

For every fixed seed $\psi \in \mathcal{P}_N$, $\left\|a_{E_R(u)\psi}\right\|_2 \le C_\psi(1+|u|)^2$ as well.

Proof. Fix $\alpha = 1/2$ in (3.10). The two-ended packet expansion used in Section 3, with two integrations by parts in its compact mass variable, yields

$$\sup_t (1+|t|)^2 \left|\sqrt{2\pi\cosh t}\,u_{n,e^{ium}f}(t,0)\right| \le C_{n,\mathrm{supp}\,f} \sum_{j=0}^{2} \left\|\partial_s^j (e^{ium(s)}f(s))\right\|_1 \le C_f(1+|u|)^2.$$

The same estimate on the bounded-t part follows from smoothness of the modes. The exponent 2 is enough because $\int (1+|t|)^{-3/2}\,\mathrm{d}t < \infty$; the radial L^2 norm is unchanged by the mass phase. Apply (3.10) and the pointwise domination following (6.5). Finite sums satisfy the triangle inequality for the orbit seminorm. □

6.2 The uncut integral and its identification

Let

$$g_{\sigma,\tau}(u) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(u-\tau)^2/(2\sigma^2)}. \tag{6.7}$$

For an arbitrary physical input $k \in \mathcal{H}_N^{\max,w}(s_*)$ and $\phi \in \mathcal{P}_{N+1}$, define directly

$$\mathcal{I}_N^+(\phi,k) = \int_{\mathbb{R}} g_{\sigma,\tau}(u)\,\mathrm{d}u \int_Y \langle \zeta_{\phi,u}(y), [R\mathsf{U}_N(u)k](y)\rangle\,\mathrm{d}y. \tag{6.8}$$

Inner products are conjugate-linear in the first entry. Both observer legs of the test and the physical input already obey the threshold. The two factors $E_R(u)$ and $\mathsf{U}_N(u)$ therefore produce $e^{-iu[m(a)-m(b)]}$ in a finite-mode matrix element, with precisely the sign in (5.3).

Theorem 6.2 (Direct creation pairing)**.** *Integral* (6.8) *is absolutely convergent after taking the norm of each pointwise Hilbert pairing. It satisfies*

$$\mathcal{I}_N^+(\phi,k)=\langle B\phi,\mathsf{C}_N k\rangle_{\mathrm{FP}}\,,\qquad |\mathcal{I}_N^+(\phi,k)|\le\sqrt{(N+1)\mathfrak{M}}\,\|B\phi\|_{\mathrm{FP}}\,\|k\|_{\mathrm{FP}}\,. \tag{6.9}$$

Here C_N *is the previously constructed Gaussian creation block at the same* σ,τ,s_**. Thus the direct formula determines the same physical output for every physical input.*

Proof. First use (2.10) and (6.6), before invoking any field:

$$\begin{aligned}\int_{\mathbb{R}} g_{\sigma,\tau}(u)\,\mathrm{d}u\int_Y \|\zeta_{\phi,u}(y)\|\,\|[R\mathsf{U}_N(u)k](y)\|\,\mathrm{d}y\\ \le C_\phi\,\|k\|_{\mathrm{FP}}\int_{\mathbb{R}} g_{\sigma,\tau}(u)(1+|u|)^2\,\mathrm{d}u<\infty.\end{aligned} \tag{6.10}$$

Jointly measurable representatives can be chosen: approximate k by physical seed images, use strong continuity of U_N, and use the continuity of R into local L^1 on each compact time interval and compact subset of Y. The resulting pairing is independent of representatives. This also proves continuity in k.

It remains to identify this independently defined integral. Take first $k=B\psi$, $\psi\in\mathcal{P}_N$, and put $\psi_u=E_R(u)\psi$. The stronger, pre-cancellation estimate is

$$\begin{aligned}\int_G \mathrm{d}g\int_Y \mathrm{d}y\;\|\zeta_{\phi,u}(y)\|\,\|[U_N(g)\psi_u](y)\| &= \int_Z a_{\zeta_{\phi,u}}a_{\psi_u}\,\mathrm{d}\mu\\ &\le C_\phi C_\psi(1+|u|)^4.\end{aligned} \tag{6.11}$$

The equality follows by the same nonnegative coarea and Haar changes of variables as (2.7). Its Gaussian integral is finite, so the u, g, and y integrals may now be interchanged. In particular the group-first integral is $R\mathsf{U}_N(u)B\psi$, not a finite-cutoff non-invariant state.

For fixed g, the u,y integral equals $\langle\phi,(1_L\otimes C_N)U_N(g)\psi\rangle_{\mathrm{kin}}$. Here is a finite-mode justification that does not assume an infinite mode interchange. The test-side annihilation in (6.3) leaves finitely many matter vectors. Only their overlaps with the transformed input matter vector contribute. Integration in θ_L selects the test's left mode, and integration in θ_R selects finitely many input observer modes by $n_b=n_a+n$. At fixed g their mass coefficients are the original compact packets times smooth principal-series matrix coefficients. Each surviving radial matrix element is therefore the absolutely convergent triple-mode integral (5.1). The Gaussian integral supplies (5.3). The same finite selection follows for the bounded C_N by approximating the transformed input in its original-mode Hilbert basis; the forbidden components have identically zero matrix elements. Thus no uniform large-angular-label bound has been assumed.

Integrating over g now gives the previously defined form (5.13), hence $\langle B\phi,\mathsf{C}_N B\psi\rangle_{\mathrm{FP}}$. Finally $B\mathcal{P}_N$ is dense, (6.10) is continuous in its physical input, and C_N is bounded. Approximation extends the identity to all k and proves the physical norm bound in (6.9). □

The identification also proves that (6.8) vanishes for a physically null test seed. Representative independence and the bound by $\|B\phi\|$ follow from the identity; they do not follow from a bound involving C_ϕ alone.

There is a useful cutoff consequence. Let $0\le\xi_j\le 1$ be compact position cutoffs tending pointwise to one. For each fixed test ϕ, the error from inserting $\xi_j(y)$ in (6.8) is at most

$$\|k\|_{\mathrm{FP}}\int_{\mathbb{R}} g_{\sigma,\tau}(u)\,\left\|a_{(1-\xi_j)\zeta_{\phi,u}}\right\|_2\,\mathrm{d}u\longrightarrow 0. \tag{6.12}$$

At fixed u this is dominated convergence in the orbit norm; the polynomial bound supplies the Gaussian majorant. A proper-time cutoff has the analogous tail bound from (6.10). Hence removal of these cutoffs converges uniformly on the physical input unit ball for a fixed test, with no exchange of conditionally convergent integrals.

6.3 What is still not a direct position theorem

Theorem 6.2 identifies the projected, time-smeared creation pairing on an explicit dense physical test family and arbitrary physical inputs. It does not assert that the raw product $\varphi^+(x_R)Rk(y)$ is a pointwise $\mathcal{K}_{N+1}$ vector, or that raw multiplication preserves observer positive frequency. The test-side finite sum is the essential domain choice.

The annihilation block is already the physical adjoint C_N^*, but its unrestricted direct position formula requires an additional argument. Transposing it would put $\mathfrak{a}_\nu^\dagger(x_R)$ on the test: even for the matter vacuum this is an infinite-mode point-field expression, so (6.4) is unavailable. Nor does the present proof remove the Gaussian, establish the $s_* = 0$ field extension, or justify integrating two physical invariant states over the full unreduced position space.

7 Annihilation as an integrated mode limit

We retain all assumptions and seed classes of Section 6. The obstruction noted there is a point-field obstruction, not yet an obstruction to convergence after integration. The following construction resolves the integrated question for every physical input. A stronger absolute position formula will then be obtained on physical seed images.

7.1 Finite emitted-mode filters

Choose finite-support real diagonal contractions on P_ν,

$$F_M|n\rangle = f_M(n)|n\rangle, \qquad 0 \le f_M(n) \le 1, \qquad f_M(n) \longrightarrow 1 \quad (n \in \mathbb{Z}). \tag{7.1}$$

Set $\mathfrak{a}_M(x) = \sum_n f_M(n)\psi_n^\nu(x)a_n$. On each fixed particle sector both this finite sum and its adjoint are bounded operators, uniformly in x for fixed M. The bound may grow with M. In particular, for $\phi \in \mathcal{P}_N$,

$$\zeta^-_{M,\phi,u}(y) = \mathfrak{a}_M^\dagger(x_R)[E_R(u)\phi](y), \qquad \left\| a_{\zeta^-_{M,\phi,u}} \right\|_2 \le C_{M,\phi}(1+|u|)^2. \tag{7.2}$$

Indeed $\|\mathfrak{a}_M^\dagger(x)v\| \le \sqrt{N+1}\,(\sum_n f_M(n)^2|\psi_n^\nu(x)|^2)^{1/2}\|v\|$; the finite-mode envelope and the packet estimate of the previous section apply. This defines, independently of a physical field operator,

$$\mathcal{I}^-_{N,M}(\phi,k) = \int_{\mathbb{R}} g_{\sigma,\tau}(u)\,\mathrm{d}u \int_Y \left\langle \zeta^-_{M,\phi,u}(y), [R\mathsf{U}_{N+1}(u)k](y) \right\rangle \mathrm{d}y, \quad k \in \mathcal{H}^{\max,w}_{N+1}(s_*). \tag{7.3}$$

Equation (2.10) proves absolute convergence and continuity in k at each fixed M. It does not supply an M-independent absolute majorant for arbitrary k.

On the kinematical right system define

$$C_{N,M} = \sqrt{N+1}(1_R \otimes P_{\mathrm{Sym}^{N+1}})(1_R \otimes F_M \otimes 1_{\text{spectators}})(T \otimes 1). \tag{7.4}$$

The filter acts on the newly emitted leg before Bose symmetrization. It is not a filter on all output particles. Thus

$$\|C_{N,M}\| \le b_N := \sqrt{(N+1)\mathfrak{M}}, \qquad C_{N,M}v \longrightarrow C_N v \quad (v \in W_N). \tag{7.5}$$

Strong convergence follows from $F_M \to 1$ strongly and boundedness of T. Since F_M commutes with rotations, $C_{N,M}$ preserves right total K. It generally fails to intertwine boosts. No physical operator descended from $C_{N,M}$, and no preservation of physical null seeds at finite M, is assumed.

7.2 Fixed-left continuity transfers the limit

On the expanded class $\mathcal{E}_N^0$ of Section 5 write $\mathcal{G}_N = J_N \mathcal{A}_N$. This agrees with B on actual seeds and satisfies

$$\|\mathcal{G}_N(|f,0\rangle_L \otimes v)\|_{\mathrm{FP}} \le B_f \|v\|, \qquad v \in W_N^{(0)}. \tag{7.6}$$

This bound concerns arbitrary right vectors of total $K = 0$; it does not require any map producing v to intertwine G.

Theorem 7.1 (Integrated annihilation limit)**.** *For every fixed $\phi \in \mathcal{P}_N$, the directly defined functionals (7.3) satisfy*

$$\sup_{\|k\|_{\mathrm{FP}} \le 1} \left| \mathcal{I}^-_{N,M}(\phi, k) - \langle B\phi, \mathsf{C}_N^* k \rangle_{\mathrm{FP}} \right| \longrightarrow 0. \tag{7.7}$$

The limit is independent of every filter sequence obeying (7.1). In particular it is the same for sharp symmetric cutoffs and Fejér weights. It is independent of physically null changes of the test seed, although an individual finite-M functional need not be.

Proof. First let $k = B\psi$, $\psi \in \mathcal{P}_{N+1}$. The joint absolute bound (6.11), now with $\zeta^-_{M,\phi,u}$, is at most $C_{M,\phi} C_\psi (1 + |u|)^4$. Hence the Gaussian permits the interchange of u, g, and y. At fixed g the emitted sum is finite. Angular integration selects only finitely many observer modes of the transformed input; its mass supports remain compact. The original local matrix element (5.1) therefore gives

$$\mathcal{I}^-_{N,M}(\phi, B\psi) = \int_G \langle (1_L \otimes C_{N,M})\phi, U_{N+1}(g)\psi \rangle_{\mathrm{kin}} \, \mathrm{d}g. \tag{7.8}$$

To check the phase, the time factors on the two sides of the local pairing give $e^{-iu(m_{\mathrm{out}} - m_{\mathrm{in}})}$. In (7.8) the creation coefficient is in the first, conjugate-linear entry, so its Gaussian phase is conjugated exactly as required for annihilation. The real Gaussian center is still τ. No time reversal or change to $-\tau$ has been made.

The first vector in (7.8) lies in $\mathcal{E}^0_{N+1}$: the left packet is unchanged and the right total magnetic label is zero. Absolute scalar group averaging on that class gives

$$\mathcal{I}^-_{N,M}(\phi, k) = \langle \kappa_M(\phi), k \rangle_{\mathrm{FP}}, \qquad \kappa_M(\phi) = \mathcal{G}_{N+1}[(1_L \otimes C_{N,M})\phi]. \tag{7.9}$$

Initially this identity holds on seed inputs. At fixed M, the left side is continuous in the physical input by (7.2); the right side is continuous by Hilbert duality. Physical density extends it to every k. In particular we have not commuted $C_{N,M}$ through a group average.

Write $\phi = \sum_{j=1}^r |f_j, 0\rangle_L \otimes v_j$. For the unfiltered intertwiner the existing descent gives $\mathcal{G}_{N+1}[(1_L \otimes C_N)\phi] = \mathsf{C}_N B\phi$. Consequently

$$\|\kappa_M(\phi) - \mathsf{C}_N B\phi\|_{\mathrm{FP}} \le \sum_{j=1}^r B_{f_j} \|(C_{N,M} - C_N)v_j\| \longrightarrow 0. \tag{7.10}$$

Equations (7.9) and (7.10) prove (7.7). Only the limiting intertwiner is used for physical descent. The bound in (7.10) is a fixed-test bound, not a uniform bound over all representatives with unit $\|B\phi\|$. □

7.3 An absolute uncut formula on seed inputs

For this stronger statement use specifically

$$f_M(n) = \left(1 - \frac{|n|}{M+1}\right)_+, \qquad K_M(\theta) = \frac{1}{M+1} \left| \sum_{j=0}^M e^{ij\theta} \right|^2. \tag{7.11}$$

The standard positive-kernel argument [9, Appendix B] extends to Hilbert-valued continuous functions. For completeness, $K_M \geq 0$, $\int K_M \, \mathrm{d}\theta/(2\pi) = 1$, and outside any fixed neighborhood of zero its integral tends to zero, since $K_M(\theta) \leq [(M+1)\sin^2(\theta/2)]^{-1}$. Uniform continuity and this bound prove uniform convergence of the convolution to a continuous periodic Hilbert-valued function. The same positivity bounds the norm of every convolution by its supremum norm.

For a finite occupation-mode vector $v \in \mathcal{K}_{N+1}$ define

$$A_{g,v}(x) = V_N(g)\mathfrak{a}_\nu(g^{-1}x)v. \tag{7.12}$$

This is a finite sum of smooth transformed scalar modes times fixed Hilbert vectors, with $\sup_{g,x} \|A_{g,v}(x)\| \leq c_v$. Covariance of the positive-frequency one-particle representation implies that its angular Fourier coefficients are those of $\sum_n \psi_n^\nu(x) a_n V_{N+1}(g)v$. This equality can first be read after smearing in x, where the annihilation operator is defined on every fixed particle sector; smoothness of (7.12) then identifies the Fourier coefficients. No point evaluation on an arbitrary one-particle Hilbert vector is needed. Fejér convolution gives

$$\mathfrak{a}_M(x)V_{N+1}(g)v \longrightarrow A_{g,v}(x), \qquad \|\mathfrak{a}_M(x)V_{N+1}(g)v\| \leq c_v. \tag{7.13}$$

Convergence is uniform in the angle for fixed g and time coordinate; no uniform convergence in g is claimed.

For a product seed $\psi(y) = u_L(x_L)u_R(x_R)v$ put $u_{R,u} = E_R(u)u_R$ and define

$$D_\psi(u;y) = \int_G u_L(g^{-1}x_L)u_{R,u}(g^{-1}x_R)A_{g,v}(x_R)\,\mathrm{d}g. \tag{7.14}$$

Extend this definition by linearity to $\mathcal{P}_{N+1}$. The scalar absolute orbit integral of the observer product is finite almost everywhere, by the seed orbit bound. Thus (7.14) is a Bochner integral almost everywhere in (u,y). For finite M, boundedness of $\mathfrak{a}_M(x_R)$ lets it pass through the seed's Bochner group average. Equation (7.13) and dominated convergence then show

$$D_\psi(u;y) = \lim_{M\to\infty} \mathfrak{a}_M(x_R)[R\mathsf{U}_{N+1}(u)B\psi](y) \quad \text{almost everywhere.} \tag{7.15}$$

Joint representatives can be chosen by the local L^1 construction in Section 6. Coarea and the two seed orbit bounds give

$$\int_{\mathbb{R}} g_{\sigma,\tau}(u)\,\mathrm{d}u \int_Y \|[E_R(u)\phi](y)\|\,\|D_\psi(u;y)\|\,\mathrm{d}y < \infty. \tag{7.16}$$

Indeed, before the g integral, the upper bound is a constant times the product of the observer seed envelopes. Its g,y integral is at most $C_{\phi,\psi}(1+|u|)^4$. This same majorant is independent of M in (7.13). We may therefore take the limit inside the direct integral and apply Theorem 7.1:

$$\int_{\mathbb{R}} g_{\sigma,\tau}(u)\,\mathrm{d}u \int_Y \langle [E_R(u)\phi](y), D_\psi(u;y)\rangle\,\mathrm{d}y = \langle B\phi, \mathsf{C}_N^* B\psi\rangle_{\mathrm{FP}}\,. \tag{7.17}$$

The pointwise Fejér limit depends only on $B\psi$: if $B\psi = 0$, every finite-M expression in (7.15) vanishes almost everywhere, and hence so does its limit.

7.4 Scope of the two conclusions

For arbitrary physical inputs the proved statement is convergence of integrated functionals in physical dual norm at each fixed seed test. For the dense physical seed images, (7.16) gives the stronger absolute uncut formula with the pointwise Fejér meaning (7.15). Neither assertion proves pointwise convergence or an absolute norm majorant for every physical input. Sharp cutoffs share the integrated limit but have not been assigned the positive majorant used for the seed formula. These distinctions complete the specified weak position interpretation of both field blocks while leaving unrestricted point multiplication, zero threshold, and removal of Gaussian smearing open.

8 Two species and ordered four-point functions

8.1 Species grading and the common domain

For two independent real scalar species use

$$\mathcal{K}_{p,q} = \mathrm{Sym}^p P_{\nu_A} \otimes \mathrm{Sym}^q P_{\nu_B}. \tag{8.1}$$

The state-space proof is unchanged: add one principal representation at a time and symmetrize only within a species. The creation bounds are $\sqrt{(p+1)\mathfrak{M}_A}$ for A and $\sqrt{(q+1)\mathfrak{M}_B}$ for B. There is no interspecies Bose projection or extra $\sqrt{2}$ in a vacuum-to-one-of-each process. The full finite-total-particle domain is invariant and consists of analytic vectors for each regulated field. Products of a finite number of these fields are therefore defined on every vector in that domain.

Bounded equivariant maps, their adjoints, and their finite compositions descend compatibly: the amplitude map satisfies $\mathcal{A}_{\mathrm{out}} C = c(d)\mathcal{A}_{\mathrm{in}}$. This also applies to actual observer spectral projections and to the time factors in (5.3). The same fixed-group-cutoff, mode-limit-first construction supplies the local weak interpretation.

Let $I = [s_*, \Lambda]$ and let E_I be the spectral projection of the actual right observer. Define the compressed field first on the common finite-particle domain by $E_I \Phi_\alpha E_I$ and then take its unique self-adjoint closure, denoted $\Phi_{\alpha,I}$. The block bounds and analytic-vector proof persist after this compression. This is more precise than assuming that an arbitrary compression of an unbounded self-adjoint operator is self-adjoint.

For a normalized matter-vacuum packet Ω_f,

$$\|\Omega_f\|^2 = \int \rho_{\mathrm{Pl}}(d)|f(d)|^2 \,\mathrm{d}d = 1, \qquad P(d) = \rho_{\mathrm{Pl}}(d)|f(d)|^2, \tag{8.2}$$

assume $\operatorname{supp} P \subset I$. Products of the compressed fields then insert E_I at every intermediate actual right-observer edge. A label restriction in one ordered CG tree is an orthogonal spectral restriction in that tree; it is not automatically the same history projection in another tree. If the external packet lies outside I, the compressed-field expression also truncates the external state and cannot be identified with an internal-only cutoff.

8.2 Noncrossed contractions

For a common width σ define

$$H_{\alpha,I}(d;u) = \int_I V_{\nu_\alpha}(a|d) e^{-\sigma^2[m(a)-m(d)]^2} e^{iu[m(a)-m(d)]} \,\mathrm{d}a. \tag{8.3}$$

Complete normalized CG orthogonality and the local scalar modulus give, with times listed from left to right,

$$\langle A_1 A_2 B_3 B_4 \rangle_f = \int P(d) H_{A,I}(d; t_1 - t_2) H_{B,I}(d; t_3 - t_4) \,\mathrm{d}d, \tag{8.4}$$

$$\langle A_1 B_2 B_3 A_4 \rangle_f = \int P(d) \,\mathrm{d}d \int_I V_{\nu_A}(a|d) e^{-\sigma^2[m(a)-m(d)]^2} e^{i(t_1-t_4)[m(a)-m(d)]} H_{B,I}(a; t_2 - t_3) \,\mathrm{d}a. \tag{8.5}$$

Here $A_j = \Phi_{A,I}(t_j)$ and $B_j = \Phi_{B,I}(t_j)$. These are complete correlators for independent species. An adjacent pair of the same species must return to the vacuum in (8.4); a nested pair yields the two successive positive weights in (8.5). They are not obtained by replacing a packet average of a product by a product of packet averages.

8.3 Crossing and its time orientation

The two ordered one-of-each states are synthesized by two different normalized CG trees. Their inner product is the corresponding S/S Racah matrix element, with the four scalar local factors. Appendix B checks its gamma normalization against the reconstructed mode vertex. Thus the zero-time crossed density agrees with (B.25)–(B.27) of [1]; for equal species labels it agrees with the assembled function (A.14).

Write that equal-label Hartle–Hawking density as

$$X_H(a,b,c,d) = a\sinh(\pi a)\, b\tanh(\pi b)\, c\sinh(\pi c)\, d\tanh(\pi d)\, \mathcal{S}^{\nu}(a,b,c,d). \tag{8.6}$$

The outer conditional density uses $\rho_H(d) = \rho_{\mathrm{Pl}}(d)\operatorname{sech}^2(\pi d)$, and the normalized external packet measure is $P(d)\,\mathrm{d}d$, not an additional Plancherel measure. For the operator order $A_1B_2A_3B_4$, the right-to-left observer path is

$$d \xrightarrow{B_4} c \xrightarrow{A_3} b \xrightarrow{B_2} a \xrightarrow{A_1} d. \tag{8.7}$$

Let $m_x = m(x)$. The exact crossed correlator of the compressed fields is

$$X_{P,I}(t_1,t_2,t_3,t_4) = \int P(d)\,\mathrm{d}d \int_{I^3} \mathrm{d}a\ \mathrm{d}b\ \mathrm{d}c\, \frac{X_H(a,b,c,d)}{\rho_H(d)} e^{-\frac{\sigma^2}{2}\mathcal{Q}}\, e^{-i\mathcal{T}}, \tag{8.8}$$

$$\mathcal{Q} = (m_d-m_a)^2 + (m_a-m_b)^2 + (m_b-m_c)^2 + (m_c-m_d)^2, \tag{8.9}$$

$$\mathcal{T} = (m_d-m_a)t_1 + (m_a-m_b)t_2 + (m_b-m_c)t_3 + (m_c-m_d)t_4. \tag{8.10}$$

At $(t,0,t,0)$ this is the archived phase $\exp[-it(m_d-m_c+m_b-m_a)]$. The half factor in each of the four Gaussian exponents is required. Coincident spectral labels are taken by the removable limits of the assembled kernel, not by evaluating divergent individual hypergeometric terms.

The operator identification follows first for compact spectral tests and finite original modes with their normalized CG transforms. Bounded creation blocks on each particle sector and the isometries $J_{p,q}$ extend it to arbitrary vacuum packets. This is sufficient for the C^1 packet below; it need not be a smooth core vector. The original unregulated spacetime-integral action on every physical distribution is not used in this argument.

For equal matter parameters and widths, set

$$u = \Phi_{A,I}(t)\Phi_{B,I}(0)\Omega_f, \qquad v = \Phi_{B,I}(0)\Phi_{A,I}(t)\Omega_f.$$

The ordered-tree norm is time independent and both norms equal $N_{P,I}$. Since $\langle v,u\rangle = X_{P,I}(t,0,t,0)$,

$$\|[\Phi_{A,I}(t),\Phi_{B,I}(0)]\Omega_f\|^2 = 2\big(N_{P,I} - \operatorname{Re} X_{P,I}(t,0,t,0)\big). \tag{8.11}$$

This concerns the specified regulated and compressed fields. No operator norm bound for a full unbounded commutator, nor a Lyapunov exponent, follows from four sampled times.

8.4 The archived finite-packet calculation

The existing calculation uses $\nu = 0.6$, $\sigma = 1$, $I = [0.2, 3]$, and

$$P(d) = \frac{8}{3w}\sin^4\!\left(\frac{\pi(d-0.7)}{w}\right)\mathbf{1}_{[0.7,0.9]}(d), \qquad w = 0.2. \tag{8.12}$$

The corresponding $f = \sqrt{P/\rho_{\mathrm{Pl}}}$ is a normalized vacuum vector. Table 1 retains numerical digits for traceability rather than claiming that many digits of integral accuracy.

The positive quantities are

$$C_{2,P,I} = 0.06733672557, \qquad D_{P,I} = 0.004543308000, \qquad N_{P,I} = 0.003396653605. \tag{8.13}$$

t	$\operatorname{Re} X_{P,I}$	$\operatorname{Im} X_{P,I}$	$2(N_{P,I} - \operatorname{Re} X_{P,I})$
0	0.003277669823	$5.05\,10^{-29}$	0.000237967563
0.5	0.003240097524	0.000166440545	0.000313112161
1	0.003127420450	0.000329368340	0.000538466309
2	0.002683193016	0.000618126682	0.001426921178

Table 1: Archived finite-packet values at $(t, 0, t, 0)$ for the fixed internal interval $[0.2, 3]$. They have empirical quadrature errors.

In particular $D_{P,I} - C_{2,P,I}^2 \simeq 9.07339\,10^{-6}$. For the single-species fourth moment at zero time, Bose symmetrization gives $D_{P,I} + N_{P,I} + \operatorname{Re} X_{P,I}(0)$.

The maximum internal quadrature refinement change was about $1.17\,10^{-6}$. The external quadrature change was smaller than that internal empirical scale and is not an independently certified error. The separate analytic bound on the crossing tail beyond spectral cap 20 is $2\,10^{-11}$, but neither the finite interval between caps 3 and 20 nor the original box quadrature has a certified total error. The operator identification does not improve those numerical error statements or turn the values in Table 1 into rigorous intervals.

9 Preparing the crossed large-mass limit

The large-mass limit must specify whether an observer threshold moves with the prepared state. We first obtain estimates independent of the large mass, then isolate the remaining local Racah limit. The latter is stated as a hypothesis, not inferred from a pointwise asymptotic formula.

9.1 Two distinct preparation protocols

Use exact observer energy $m(s) = \sqrt{s^2 + 1/4}$, its inverse $s(m) = \sqrt{m^2 - 1/4}$, and $j(m) = m/s(m)$. Let $\pi_0(r)$ be a fixed probability density on $r \geq 0$. A translated normalized vacuum packet is specified by

$$\rho_{\mathrm{P1}}(d)|f_M(d)|^2\,\mathrm{d}d = \pi_0(r)\,\mathrm{d}r, \qquad m(d) = M + r. \tag{9.1}$$

The phase of f_M is immaterial to these pure-right vacuum correlators. The earlier flat clock variable is $p = 2\pi r$; if its density is $P(p)$, then $\pi_0(r) = 2\pi P(2\pi r)$. We distinguish two compressions applied at every insertion:

$$Q^{\mathrm{fixed}} = \mathbf{1}_{H_R \geq m_*}, \quad m_* > 1/2; \qquad Q_M^{\mathrm{moving}} = \mathbf{1}_{H_R \geq M}, \quad M \geq 1. \tag{9.2}$$

The left observer is already in the support of the packet. Its matching threshold projection acts trivially throughout a pure-right correlator. In excess energy the first boundary retreats to $m_* - M$, whereas the second remains at zero. The fixed-threshold problem retains our positive s_* assumption and does not assert the $s_* = 0$ field extension.

The microcanonical preparation of [1, (6.29), (B.43)] has a window on the external state, without a moving threshold at every intermediate insertion. Its variables $s_i = \Lambda + q_i$ are representation labels; our $M + r_i$ are exact energies. On bounded offsets, differences of the two labels agree with energy differences up to $O(M^{-2})$. The paper's fixed-time delta limit and its scrambling-time limit are different distributional limits. Its two time origins are $\log(2\Lambda)$ in (6.32) and $\log(2\Lambda/e)$ in (B.45). Neither distinction is removed by a normalization convention for our packet.

9.2 Uniform tails without an absolute bound on the Racah kernel

The one-step positive density in exact energy is

$$K(m'|m) = j(m')V_\nu(s(m')|s(m))e^{-\sigma^2(m'-m)^2}. \tag{9.3}$$

Fix $\nu, \sigma > 0$ and a common lower mass $m_* > 1/2$ for all states under consideration. Equation (5.10) and $j(m') \le j(m_*)$ give a constant C such that

$$0 \le K(m'|m) \le B(m'-m) := Ce^{-\sigma^2(m'-m)^2}. \quad A := \int_{\mathbb{R}} B(u)\,\mathrm{d}u < \infty. \tag{9.4}$$

Additional intermediate projections only reduce positive norms. For a vacuum fiber at energy m_0, the squared norm of either ordered one-of-each state is the integral of $K(m_1|m_0)K(m_2|m_1)$ and is at most A^2. This uses the normalized ordered CG tree, not orthogonality of arbitrary noncommuting projection histories.

Proposition 9.1 (Mass-independent spectral-tail control)**.** *In each ordered tree retain only $|m_1 - m_0| \le R$ and $|m_2 - m_0| \le R$. Let $X_{M,R}$ pair the two retained trees, and let X_M be the uncut crossed pairing. Define $b(R) = \int_{|u|>R} B(u)\,\mathrm{d}u$. For either protocol in (9.2), any normalized packet, and all real times,*

$$|X_M - X_{M,R}| \le 2A\sqrt{2A\,b(R/2)}. \tag{9.5}$$

Discarding additionally the external packet tail $r > L$ changes the crossed pairing by at most $A^2 \int_L^\infty \pi_0(r)\,\mathrm{d}r$.

Proof. Write $u = m_1 - m_0$, $v = m_2 - m_1$. Leaving the retained path implies $|u| > R/2$ or $|v| > R/2$. Ordered-channel orthogonality identifies its squared norm with the positive excluded path integral. Thus it is at most $2Ab(R/2)$, uniformly in m_0 and time. Subtract the two pairings and apply Cauchy–Schwarz, using norm at most A for each tree. The fiber bound $|X_{m_0}| \le A^2$ gives the external-tail assertion. □

The cutoff in this proposition is a numerical path restriction centered on the external energy. It is not an additional fixed experimental mass window. In particular, the old box $[0.2, 3]$ cannot follow $M \to \infty$. The constants are analytic finite constants; no certified numerical value of C is claimed.

9.3 The unresolved compact distribution limit

Let $\mathcal{C}_\nu(a,b,c;d) = X_H(a,b,c,d)/\rho_H(d)$ with exactly the normalization in Section 8. In the variables

$$d = s(M+r), \quad a = s(M+r+x), \quad b = s(M+r+y), \quad c = s(M+r+z), \tag{9.6}$$

define the density relative to $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ by

$$\widetilde{\mathcal{C}}_M(r;x,y,z) = j(M+r+x)j(M+r+y)j(M+r+z)\mathcal{C}_\nu(a,b,c;d). \tag{9.7}$$

All masses in a compact set of offsets are positive for sufficiently large M. The natural local hypothesis is

$$\widetilde{\mathcal{C}}_M \longrightarrow v_\nu(x)v_\nu(z)\delta(y-x-z) \quad \text{in } \mathcal{D}'(\mathbb{R}^4_{r,x,y,z}), \tag{9.8}$$

where

$$v_\nu(u) = \frac{e^{-\pi u}}{16\pi^3}\left|\Gamma\left(\tfrac{1}{4} + \tfrac{i}{2}(\nu+u)\right)\Gamma\left(\tfrac{1}{4} + \tfrac{i}{2}(\nu-u)\right)\right|^2, \qquad k(u) = v_\nu(u)e^{-\sigma^2u^2}. \tag{9.9}$$

The normalization is consistent with the assembled factors (B.41), (B.42), (B.46) of [1]: their conditional density on the delta surface has coefficient $e^{-\pi(x+z)}\Gamma_4(\nu,x)\Gamma_4(\nu,z)/(256\pi^6)$. This coefficient check is not a proof of (9.8).

There is a specific unresolved analytic step. The exact ${}_4F_3(1)$ series has excess one. The ${}_2F_1(1)$ appearing in (B.33) has excess $i(s_1-s_2+s_3-s_4)$, with zero real part. Its gamma expression therefore involves continuation, not ordinary absolute summation at one. Uniform control of the four combined branches near that excess being zero is required. The fixed-label simplification alone does not bound the remainder against a compact smooth test, especially near the delta surface and coincident-label charts.

Proposition 9.2 (Conditional reduction to a finite-energy problem)**.** *If* (9.8) *holds, the exact crossed correlators of both protocols converge for every fixed probability density* π_0*. The convergence is uniform on compact sets of the four real times. For the moving threshold the limit is*

$$\begin{aligned} X_\infty(\mathbf{t}) &= \int_{\mathbb{R}^2} k(x)k(z)K_0((-x)_+ + (-z)_+)e^{ix(t_1-t_3)+iz(t_2-t_4)}\,\mathrm{d}x\,\mathrm{d}z, \\ K_0(q) &= \int_q^\infty \pi_0(r)\,\mathrm{d}r, \qquad q \geq 0. \end{aligned} \tag{9.10}$$

For the fixed threshold, K_0 *in this expression is replaced by one. No rate or logarithmically growing time interval follows from this proposition.*

Proof. First truncate external r and path offsets by Proposition 9.1. Approximate π_0 in L^1 by smooth compact densities; the fiber norm bound controls the error by A^2 times that L^1 error. For the moving boundary, replace each intermediate hard step by a smooth step differing only in an energy layer of width η. The excluded norm squared is at most $2\eta A \sup B$ per tree, because either the first or the final energy is in the layer. Thus the pairing error is $O(\sqrt{\eta})$ uniformly in M. External layers are controlled by the mass of π_0 in them. Smooth cutoffs on the retained offsets can similarly be chosen, or the outer cutoff can be enlarged. We now have a compact smooth test for (9.8).

The exact Gaussian and time factors in these coordinates are

$$\begin{aligned} \mathcal{Q} &= x^2 + (y-x)^2 + (y-z)^2 + z^2, \\ \mathcal{T} &= -xt_1 + (x-y)t_2 + (y-z)t_3 + zt_4. \end{aligned} \tag{9.11}$$

On $y = x+z$ they give $k(x)k(z)$ and the phase in (9.10). The four surviving moving-threshold conditions are $r, r+x, r+z, r+x+z \geq 0$, equivalent to $r \geq (-x)_+ + (-z)_+$. The fixed boundary eventually leaves any compact set of offsets, and gives no such condition. Remove the smooth layers, then the truncations, using the uniform bounds and their limiting positive-weight counterparts.

Finally, differentiating a tree's time phase introduces one of its energy increments. Equation (9.4) has a finite second increment moment, so the tree derivatives and hence the four-point Lipschitz constants are bounded independently of M. A finite time net upgrades pointwise convergence to uniform convergence on compact time sets. This argument supplies no growing time net with controlled convergence rate. □

9.4 A boundary contribution in the comparison model

Independently of the unproved hypothesis, define a flat-energy comparison model with a clock on $L^2(\mathbb{R}_+)$ and two independent particle species labelled by increments $x \in \mathbb{R}$. An emission has amplitude $\sqrt{k(x)}$, changes clock energy from r to $r+x$, and is projected onto $r+x \geq 0$. Negative particle labels here encode frequencies; they are not an assumed physical Hamiltonian

spectrum of de Sitter matter. On the vacuum and one-particle sectors this construction is bounded since $k \in L^1$. Smearing times multiply each emission by e^{-ixt}. It is a comparison of ordered amplitudes, not a constructed full de Sitter dual.

For equal species parameters the two ordered norms and their crossed overlap at $(t, 0, t, 0)$ are, respectively,

$$N_\infty = \int k(x)k(z)K_0(\max(0, -x, -x - z))\,\mathrm{d}x\,\mathrm{d}z,$$
$$X_\infty = \int k(x)k(z)K_0((-x)_+ + (-z)_+)\,\mathrm{d}x\,\mathrm{d}z. \tag{9.12}$$

These are direct formulas in this explicitly defined model. The second formula is also the conditional prediction of Proposition 9.2 for the exact problem. Both are independent of t at this repeated-time configuration. Subtracting the two ordering indicators gives the positive identity

$$\mathcal{D}_{\mathrm{edge}} := 2(N_\infty - X_\infty) = 2\int_0^\infty \pi_0(r)\,\mathrm{d}r \int_0^\infty k(x)\,\mathrm{d}x \int_r^{r+x} k(-u)\,\mathrm{d}u > 0. \tag{9.13}$$

Indeed the only unequal region is an energy gain $x > 0$ followed by a loss $u > r$, while $u \le r + x$ allows the final clock state. The reverse order is removed at its first step. Since k is strictly positive at all finite real increments, the inner integrals are positive for every finite r, proving strict positivity for any probability density. This is the commutator norm squared in the comparison model; its identification with a large-mass limit of the exact commutator remains conditional on (9.8).

If a fixed packet is shifted by an additional excess energy $h \to \infty$, dominated convergence in (9.13) gives $\mathcal{D}_{\mathrm{edge}} \to 0$. Removing the clock boundary altogether gives $N_\infty = X_\infty = (\int k)^2$. A nonzero repeated-time commutator for a moving threshold can therefore include a static projection contribution. It cannot by itself diagnose recoil growth or a Lyapunov exponent. Establishing the local distribution limit and then its time-dependent remainder is the next exact-model task.

10 Resolving the hypergeometric endpoint before assembling the kernel

We now give an exact integral representation and a two-term asymptotic for each balanced hypergeometric factor. This is a special-function step towards (9.8), not a proof of that hypothesis. The four beta-like prefactors in (B.21)–(B.24) of [1] still have to be combined, particularly on their removable-pole surfaces.

10.1 An exact Euler representation

Write a factor, after permuting its parameters, as

$$H_\Lambda = {}_4F_3(A, C, b, d; E, F, g; 1), \quad A = W + A_0, \quad C = W + C_0, \quad W = \pm 2i\Lambda, \tag{10.1}$$

where $E = A + \alpha$, $F = C + \beta$, $g = b + d + \delta$, and $\alpha + \beta + \delta = 1$. All of $A_0, C_0, b, d, \alpha, \beta$ have real part $1/2$ in the real-label problem; δ is purely imaginary. The offsets range over a fixed compact set and $\Lambda > 0$. Let B denote Euler's beta function, not the function in KL (B.23). Twice applying the Euler transform [6, 16.5.2] gives

$$H_\Lambda = \mathcal{N}_\Lambda \int_0^1 \int_0^1 u^{A-1}(1-u)^{\alpha-1} v^{C-1}(1-v)^{\beta-1}\, {}_2F_1(b, d; g; uv)\,\mathrm{d}u\,\mathrm{d}v,$$
$$\mathcal{N}_\Lambda = \frac{\Gamma(E)\Gamma(F)}{\Gamma(A)\Gamma(C)\Gamma(\alpha)\Gamma(\beta)}. \tag{10.2}$$

The identity includes the argument one. One may first insert an argument $0 < z < 1$ and pass to $z = 1$. The Euler representation of the Gauss function has a third weight with real exponents $-1/2$, and an additional factor $(1 - zuvt)^{-b}$. Near $u = v = t = 1$ its modulus is bounded by a constant times $[\xi\eta\zeta(\xi + \eta + \zeta)]^{-1/2}$, where $\xi = 1 - u$, $\eta = 1 - v$, $\zeta = 1 - t$. This is integrable in three dimensions. All other faces are integrable beta endpoints. Dominated convergence therefore justifies the endpoint identity without using a divergent Gauss series at one.

For $\delta \neq 0$, the ordinary, unregularized connection formula is

$$\begin{aligned} {}_2F_1(b, d; g; z) &= G_+ \, {}_2F_1(b, d; 1 - \delta; 1 - z) \\ &\quad + G_-(1 - z)^{\delta} {}_2F_1(g - b, g - d; 1 + \delta; 1 - z), \\ G_+ &= \frac{\Gamma(g)\Gamma(\delta)}{\Gamma(g - b)\Gamma(g - d)}, \qquad G_- = \frac{\Gamma(g)\Gamma(-\delta)}{\Gamma(b)\Gamma(d)}. \end{aligned} \tag{10.3}$$

This follows from [6, 15.8.4] after restoring $\Gamma(g)$. All powers here use the real logarithm for $0 < z < 1$.

10.2 A uniform two-branch expansion

Proposition 10.1. *For the compact real-offset families specified above, and any fixed $0 < \epsilon < 1/4$,*

$$H_\Lambda = G_+ + \frac{G_-}{\Gamma(1 - \delta)} W^{-\delta} + O(\Lambda^{-1+\epsilon}). \tag{10.4}$$

The error is uniform, including at $\delta = 0$, where the sum is interpreted by its removable limit. With $\log W = \log(2\Lambda) \pm i\pi/2$, that limit is

$$\frac{\Gamma(b + d)}{\Gamma(b)\Gamma(d)} [\log W + \psi(1) - \psi(b) - \psi(d)]. \tag{10.5}$$

No uniform estimate for the complete Racah kernel is asserted here.

Proof. Set $u = e^{-p}$, $v = e^{-q}$, $w = p + q$, $t = p/w$, and $a = \alpha + \beta = 1 - \delta$. Equation (10.2) becomes

$$H_\Lambda = \mathcal{N}_\Lambda \int_0^\infty e^{-Ww} w^{a-1} J(w) \, {}_2F_1(b, d; g; e^{-w}) \, \mathrm{d}w, \tag{10.6}$$

where, putting $h(z) = (1 - e^{-z})/z$ with $h(0) = 1$,

$$\begin{aligned} J(w) = \int_0^1 & t^{\alpha-1}(1 - t)^{\beta-1} e^{-w[A_0 t + C_0(1-t)]} \\ & \times h(wt)^{\alpha-1} h(w(1 - t))^{\beta-1} \, \mathrm{d}t. \end{aligned} \tag{10.7}$$

This is smooth near zero, uniformly on the compact parameter sets, and $J(0) = \mathrm{B}(\alpha, \beta)$. At large w the integrand amplitude and all its derivatives decay exponentially times a polynomial. This follows from $\Re A_0, \Re C_0 > 0$, the behavior $h(z) \sim z^{-1}$, and the convergent Gauss series at $e^{-w} = 0$.

Away from $\delta = 0$, the connection formula splits the amplitude at zero into $G_+ w^{a-1}$ and $G_- w^{a+\delta-1} = G_-$, each multiplied by a smooth factor with value $\mathrm{B}(\alpha, \beta)$. A smooth cutoff at zero and the Abel boundary value of the Laplace integral give the leading Fourier integrals $\Gamma(a)W^{-a}$ and W^{-1}. The remainders have powers w^a and w with bounded differentiated amplitudes. Splitting at $w = |W|^{-1}$ and integrating by parts on dyadic annuli (more than three times if needed) bounds their transforms by $O(|W|^{-\Re a - 1})$ and $O(|W|^{-2})$, respectively. The part away from zero is rapidly decreasing by integration by parts.

Uniform gamma-ratio asymptotics give

$$\mathcal{N}_\Lambda \mathrm{B}(\alpha,\beta) = \frac{W^a}{\Gamma(a)}[1+O(\Lambda^{-1})].$$

Consequently the two leading terms are those in (10.4). On compact complex parameter sets with $|\Re\delta| \le \epsilon$ and separated from zero, the error is $O(\Lambda^{-1+\epsilon})$; the second term's modulus is at most a constant times Λ^ϵ.

To cover coalescence, keep A_0, C_0, b, d, α fixed and vary δ holomorphically, setting $\beta = 1-\alpha-\delta$ and $g = b+d+\delta$. On $|\delta| = \epsilon$ all Euler endpoint real parts remain positive. Both H_Λ and the combined leading expression are holomorphic inside this circle. Indeed $\Gamma(-\delta)/\Gamma(1-\delta) = -1/\delta$ makes the cancellation explicit:

$$G_+ + \frac{G_- W^{-\delta}}{\Gamma(1-\delta)} = \frac{\Gamma(g)}{\Gamma(b)\Gamma(d)} \frac{\dfrac{\Gamma(1+\delta)\Gamma(b)\Gamma(d)}{\Gamma(b+\delta)\Gamma(d+\delta)} - e^{-\delta\log W}}{\delta}. \tag{10.8}$$

The maximum-modulus principle transfers the boundary error estimate to the disk. Expanding its numerator at zero gives (10.5). A finite compact covering completes uniformity. □

For KL (B.33), with $s_i = \Lambda + q_i$ and $s_A = s_B = \nu$, one choice is

$$\begin{aligned} A_0 &= \tfrac12 - i(q_3+q_4+\nu), & C_0 &= \tfrac12 - i(q_2+q_3-\nu),\\ b &= \tfrac12 + i(q_4-q_3-\nu), & d &= \tfrac12 + i(q_2-q_3+\nu),\\ \alpha &= \tfrac12 + i(q_4-q_3+\nu), & \beta &= \tfrac12 + i(q_2-q_1-\nu),\\ \delta &= i(q_1-q_2+q_3-q_4), & W &= -2i\Lambda. \end{aligned} \tag{10.9}$$

The other three factors in (B.21)–(B.24) have the same structure after permutations, with either sign of W. For a fixed nonzero $\delta = i\chi$, $|W^{-\delta}| = e^{\pm\pi\chi/2}$ is independent of Λ. Thus a single hypergeometric factor cannot generally be replaced pointwise by G_+ alone.

This does not refute the assembled distributional asymptotic in KL: the extra terms may cancel between the four branches. Our finite-point diagnostics show substantial cancellation at one generic offset tuple, but do not prove it or cover the opposite-label coincidence surfaces. The next step is to multiply by the exact beta-like prefactors, prove the cancellation of the additional leading terms, and bound the combined remainder across those surfaces before applying the spectral-tail result.

11 Assembly and the fixed-time crossed limit

This section completes the compact distribution step for the specified equal-species, strict-$G_N \to 0$ model. We use the exact four terms in KL (B.21)–(B.24), rather than dropping the second endpoint branch of an individual hypergeometric function. The delta coefficient agrees with KL (B.46); the purpose here is to control the combined remainder and the finite-packet limit. No originality claim or finite-G_N result is intended.

11.1 Finite parameters and the four coefficients

Put $s_j = \Lambda + q_j$, with real offsets in a fixed compact set, and define

$$\begin{aligned} p &= \tfrac12 + \tfrac{i}{2}(q_4-q_2-2\nu), & q &= \tfrac12 + \tfrac{i}{2}(q_4-q_2+2\nu),\\ e &= i(q_1-q_3), & \delta &= i(q_1-q_2+q_3-q_4),\\ t &= (\delta-e)/2, & v &= (\delta+e)/2. \end{aligned} \tag{11.1}$$

Here the scalar q is distinguished from the four offsets q_j. Write

$$c(z) = \sqrt{\pi}\,\frac{\Gamma((1-z)/2)}{\Gamma(z/2)}. \qquad c(z)c(1-z) = \pi. \tag{11.2}$$

The beta-like function of KL (B.23) is exactly $B(u,w) = c(u)c(w)/c(u+w-1)$. Uniform gamma-ratio estimates give

$$B(u, \pm 2i\Lambda + w_0) = c(u)\Lambda^{u-1}[1 + O(\Lambda^{-1})] \tag{11.3}$$

on compact offset sets away from the indicated meromorphic poles. There is no arbitrary constant in the phase of this formula: the powers come from the principal logarithms of the two opposite large arguments.

Set $D = c(1-p-t)c(q+v)c(1-q+t)c(p-v)$. Denote the exact prefactors of the four hypergeometric functions by $P_{j,\Lambda}$, in the order of the two terms of the first G, then the two terms of the second G in KL (B.21). Thus $\pi^3 S = \sum_j P_{j,\Lambda} H_{j,\Lambda}$. The complete parameter and leading-prefactor table is

j	b_j	d_j	g_j	δ_j	W_j
1	$1-p-v$	$p-v$	$1-e$	δ	$-2i\Lambda$
2	$1-q+t$	$q+t$	$1-e$	$-\delta$	$2i\Lambda$
3	$1-q+v$	$q+v$	$1+e$	$-\delta$	$2i\Lambda$
4	$1-p-t$	$p-t$	$1+e$	δ	$-2i\Lambda$

$$\begin{aligned} P_{j,\Lambda} &= \Lambda^{\delta_j - 1} P_j^0 [1 + O(\Lambda^{-1})], \\ (P_1^0, P_2^0, P_3^0, P_4^0) &= D\left(\frac{c(p+v)}{c(p-v)c(e)}, \frac{c(1-q-t)}{c(1-q+t)c(e)}, \frac{c(q-v)}{c(q+v)c(-e)}, \frac{c(1-p+t)}{c(1-p-t)c(-e)}\right). \end{aligned} \tag{11.4}$$

For example, write the original h variables as $h_1 = p$, $h_3 = q$, $h_2 = H$, $h_4 = 1-H$, $h = K$, $h' = L$. Then $H-K = t$, $H-L = v$, $K-L = e$. The first prefactor consists of $B(1+H-K-q, 2-2H)$, $B(1+K-p-H, p+H-L)$, and $B(H+q-L, K+H-q)$. Applying (11.3) to the first and third yields P_1^0. The second term follows by the other choice of h_4, and the second G by $(p,q,H,K,L) \mapsto (q,p,1-H,1-K,1-L)$. This determines the table without choosing branches by numerical size.

11.2 Cancellation and a pole-free leading expression

The additional endpoint branch of term j, after multiplying by its leading prefactor, has coefficient

$$E_j = -\frac{P_j^0 \Gamma(g_j)}{\delta_j \Gamma(b_j)\Gamma(d_j)} (W_j/\Lambda)^{-\delta_j} \tag{11.5}$$

in $\Lambda\pi^3 S$. The identity $\sum_j E_j = 0$ is exact as a meromorphic identity. To check it explicitly, put $P = \pi p/2$, $Q = \pi q/2$, $T = \pi t/2$, $V = \pi v/2$. Using $c(z) = 2^z \Gamma(1-z)\sin(\pi z/2)$, their sum is $2De^{i\pi\delta/2}/[\pi\delta\sin(\pi e/2)]$ times

$$\begin{aligned} &-\sin(P+V)\cos(P-V) + \cos(Q+T)\sin(Q-T) \\ &\quad -\sin(Q-V)\cos(Q+V) + \cos(P-T)\sin(P+T) = 0. \end{aligned} \tag{11.6}$$

The equality follows by writing each product as a sum of two sines.

The remaining coefficients also simplify. Define

$$U_+ = 2^{1-\delta}\pi\cos(\pi\delta/2)\frac{c(p+v)c(q+v)}{c(p-t)c(q-t)},$$
$$U_- = 2^{1+\delta}\pi\cos(\pi\delta/2)\frac{c(p-v)c(q-v)}{c(p+t)c(q+t)}. \tag{11.7}$$

They equal, respectively, $\sum_{j=1,4} P_j^0\Gamma(g_j)/[\Gamma(g_j-b_j)\Gamma(g_j-d_j)]$ and the analogous sum over $j=2,3$. One way to verify the first identity is to factor $D/[c(p-v)c(p-t)\Gamma(p+t)\Gamma(p+v)]$ from the two terms. The remaining numerator is

$$\sin(P-T)\cos(P+T)-\sin(P-V)\cos(P+V)=\cos(\pi\delta/2)\sin(\pi e/2),$$

which cancels the apparent e pole. The second identity follows by the same interchange as above. In particular,

$$U_+|_{\delta=0} = U_-|_{\delta=0} = 2\pi \tag{11.8}$$

for every allowed p, q, e, including $e=0$.

Theorem 11.1 (Uniform assembled remainder)**.** *On compact real-offset sets, with fixed $\nu>0$, for any $0<\epsilon<1/4$,*

$$S(\Lambda+q_1,\dots,\Lambda+q_4,\nu,\nu) = S_\Lambda^0 + O(\Lambda^{-2+\epsilon}),$$
$$S_\Lambda^0 = \frac{1}{\pi^3\Lambda}[\Lambda^\delta\Gamma(\delta)U_+ + \Lambda^{-\delta}\Gamma(-\delta)U_-]. \tag{11.9}$$

The bracket has its removable value at $\delta=0$; the estimate includes both opposite-label coincidence surfaces and their intersection.

Proof. First stay away from $e=0$ and $\delta=0$. Multiplying the endpoint expansion of Proposition 10.1 by (11.4), and using the exact cancellation above, gives the displayed remainder. To pass through the apparent poles, we require estimates in a small complex tube, not just continuity on the real labels.

The Euler/Fourier proof extends to such tubes: the real parts of $A_0, C_0, \alpha, \beta, b, d, g-d$ remain positive, the balanced identity $\alpha+\beta+\delta=1$ remains exact, and the triple-corner integrability is unchanged under sufficiently small perturbations. On compact subsets of this tube separated from $\delta=0$, that proof gives a remainder $O(\Lambda^{-1+\eta})$ when $|\Re\delta|\le\eta$. The prefactors have size $O(\Lambda^{-1+\eta})$ off $e=0$, while their relative gamma-ratio error is $O(\Lambda^{-1})$. Thus the combined remainder is $O(\Lambda^{-2+2\eta})$ there. Choose $2\eta<\epsilon$.

The exact S is holomorphic in a mass-independent small tube about any compact real-offset set, by its real triple integral KL (B.10)–(B.14). For completeness, use signed logarithmic coordinates for its three integration variables. The real-label envelope is an exponential decay times the four hyperplane singularities $|u|^{-1/2}$, $|v|^{-1/2}$, $|w|^{-1/2}$, $|u+v+w|^{-1/2}$. A small imaginary perturbation of the labels weakens the exponential rate and changes each singular exponent by at most $C\eta$. Choose the tube so that each exponent is below one and their sum at the common origin is below three, while the exponential rate stays positive. The same envelope with these margins dominates parameter derivatives, proving holomorphy. The large real mass appears only in oscillatory phases and does not shrink this tube.

The explicit $U_\pm$ are holomorphic throughout the tube: their c arguments remain near real part $1/2$. Equation (11.8) removes the δ pole, and there is no e denominator. Hence $S-S_\Lambda^0$ is holomorphic in e and δ across the relevant surfaces. Apply the maximum-modulus principle on small disks in these variables, using a product of circles near their intersection. The boundary estimates just obtained transfer to the interiors. Compact covering gives the asserted real-parameter estimate. In this chart only $e=0$ is an apparent prefactor pole; the other opposite-label surface introduces none, so is already included. □

For numerical evaluation at $\delta = 0$, define $\ell(z) = \partial_z \log c(z) = -[\psi((1-z)/2) + \psi(z/2)]/2$. Then

$$S_\Lambda^0|_{\delta=0} = \frac{4}{\pi^2 \Lambda}\left[\log(\Lambda/2) - \gamma_E + \frac{1}{2}\sum_{z=p,q}\{\ell(z+e/2) + \ell(z-e/2)\}\right]. \tag{11.10}$$

No coincident-label perturbation is needed in this formula.

11.3 The local distribution and exact-energy variables

Write $\chi = q_1 - q_2 + q_3 - q_4$, so $\delta = i\chi$. Equation (11.8) isolates the singular part of ΛS_Λ^0 as

$$\frac{4}{\pi^2}\frac{\sin(\chi \log \Lambda)}{\chi}. \tag{11.11}$$

After subtracting this expression, the two remaining amplitudes are smooth through $\chi = 0$ and multiply $e^{\pm i\chi \log \Lambda}$. Their pairing with a compact smooth test tends to zero by integration by parts. The sinc distribution therefore proves

$$\Lambda S \longrightarrow \frac{4}{\pi}\delta(\chi) \quad \text{in distributions on the four offsets.} \tag{11.12}$$

This also shows $\Lambda S = O(1 + \log \Lambda)$ pointwise on compact real offset sets, a useful bound when changing coordinates.

Return to the ordering (a, b, c, d) used in our conditional kernel and write $\mathcal{C}_\nu = F_\Lambda S$. Directly from A.14 and Section 8,

$$F_\Lambda = \frac{abc \sinh(\pi a)\tanh(\pi b)\sinh(\pi c)\cosh^2(\pi d)}{1024\pi^9}\,\Gamma_{16}(a, b, c, d). \tag{11.13}$$

Here Γ_{16} is exactly the sixteen-gamma product of A.14. Its eight large arguments give $16\pi^4\Lambda^{-2}e^{-\pi(a+b+c+d)}[1 + O(\Lambda^{-1})]$; the remaining eight factors, denoted Γ_{diff}, depend only on adjacent label differences. Consequently

$$\frac{F_\Lambda}{\Lambda} \longrightarrow \frac{e^{\pi(d-b)}}{1024\pi^5}\Gamma_{\text{diff}} \tag{11.14}$$

smoothly on compact offsets, with an $O(\Lambda^{-1})$ remainder in every fixed derivative order. The phases $\Lambda^{i\eta\nu}$ of the four large gamma pairs cancel because $\sum \eta = 0$.

On $a = d + x$, $c = d + z$, $b = d + x + z$, multiplication of (11.12) by (11.14) gives

$$\frac{e^{-\pi(x+z)}}{256\pi^6}\prod_{u=x,z}\left|\Gamma\left(\tfrac{1}{4} + \tfrac{i}{2}(\nu + u)\right)\Gamma\left(\tfrac{1}{4} + \tfrac{i}{2}(\nu - u)\right)\right|^2 = v_\nu(x)v_\nu(z). \tag{11.15}$$

The two orderings of χ differ by a sign only, leaving the delta measure unchanged.

Finally, the exact-energy change $s(M + \xi) - M = \xi + O(M^{-1})$ is a smooth diffeomorphism close to the identity on every compact set. In a four-variable test pairing, its Jacobian cancels the three internal factors in (9.7), leaving only the inverse external Jacobian, which tends smoothly to one. The transformed test differs by $O(M^{-1})$ in every fixed smooth seminorm. The compact $O(1 + \log M)$ bound just proved makes this difference vanish even against the moving kernel. Thus we obtain the formerly conditional step:

Theorem 11.2 (Compact limit and finite-packet consequence)**.** *For the model and normalizations of Sections 8 and 9, equation (9.8) holds. Proposition 9.2 therefore applies without its extra hypothesis: both threshold protocols have their stated fixed-time limits, uniformly on compact sets of real times, for every fixed probability density $\pi_0 \in L^1(\mathbb{R}_+)$.*

The moving-threshold commutator limit is the positive static term (9.13): the ordered norms converge by the one-step gamma-ratio limit and the same Gaussian domination. For a fixed threshold the repeated-time commutator limit is zero. These statements retain the fixed Gaussian width and strict $G_N \to 0$. They do not determine the scrambling-time remainder, a Lyapunov exponent, or a finite-G_N holographic dual.

12 The logarithmic-time limit with finite preparation

We keep the state, group, equal independent matter species, exact energy, fixed proper Gaussian width and the two threshold protocols of Section 9. The limit is still strictly $G_N \to 0$. The result below is a distribution and packet limit in this model, not a finite-gravity scrambling theorem. Throughout this section the repeated times are $(t_M, 0, t_M, 0)$, with

$$t_M = \log(2M) + T, \qquad T \in \mathbb{R}. \tag{12.1}$$

The order of limits in the early and late plateaux below is first $M \to \infty$ at fixed T, and only then $T \to \mp\infty$.

12.1 Phase orientation and the surviving distribution

For representation-label offsets write $\chi = q_1 - q_2 + q_3 - q_4$, $\delta = i\chi$, and use $p, q, e, t, v, U_\pm$ from Section 11; the auxiliary scalar t there is not the physical time t_M. Define

$$V_+ = 2^{-\delta} U_+, \qquad V_- = 2^{\delta} U_-. \tag{12.2}$$

Both have value 2π on $\chi = 0$. The leading expression (11.9) becomes

$$\Lambda S^0_\Lambda = \frac{1}{\pi^3}\left[(2\Lambda)^{i\chi}\Gamma(i\chi)V_+ + (2\Lambda)^{-i\chi}\Gamma(-i\chi)V_-\right]. \tag{12.3}$$

Each summand on the right is assigned its symmetric principal value in χ; their sum is the original removable, ordinary function. This is an algebraic decomposition, not an arbitrary choice of contact term.

In our ordering (a, b, c, d) the repeated-time phase is $e^{it_M\chi_m}$, where $\chi_m = m_a - m_b + m_c - m_d$; see (8.10). The minus branch, rather than the plus branch, therefore survives positive logarithmic time. Set

$$\mathcal{L}_T(q_1, q_2, q_3, q_4) = \frac{2}{\pi}\delta_D(\chi) + \frac{1}{\pi^3}\,\mathrm{PV}_\chi\left[e^{iT\chi}\Gamma(-i\chi)V_-\right]. \tag{12.4}$$

Here PV_χ means symmetric exclusion of $|\chi| < h$ followed by $h \downarrow 0$, at fixed three transverse linear coordinates. The transverse choice does not alter this linear-hyperplane distribution.

Theorem 12.1 (Local logarithmic-time limit)**.** *On compact real-offset tests, for fixed $\nu > 0$,*

$$\Lambda e^{i[\log(2\Lambda)+T]\chi} S(\Lambda + q_1, \ldots, \Lambda + q_4, \nu, \nu) \longrightarrow \mathcal{L}_T. \tag{12.5}$$

Convergence is uniform for T in compact real intervals.

Proof. The remainder in Theorem 11.1, after multiplication by Λ and a unit-modulus phase, is $O(\Lambda^{-1+\epsilon})$ on each compact set. Thus it remains to handle (12.3). Locally,

$$\Gamma(i\chi)V_+ = -\frac{2\pi i}{\chi} + R_+, \qquad \Gamma(-i\chi)V_- = \frac{2\pi i}{\chi} + R_-,$$

with smooth $R_\pm$. Multiplication of the plus branch by the time phase gives frequency $2\log(2\Lambda)+T$. The Fourier identity

$$e^{ik\chi}\,\mathrm{PV}\,\frac{1}{\chi}\longrightarrow i\pi\,\mathrm{sgn}(k)\delta_D(\chi),\qquad |k|\to\infty,$$

and integration by parts for the smooth remainder give the positive $2\delta_D(\chi)/\pi$ term. The minus branch has only the finite phase $e^{iT\chi}$. These arguments are uniform on compact T intervals: the large frequency remains positive and the finite-phase test seminorms are bounded there. Smooth compact transverse integrations cause no change. □

12.2 Restoring external factors and exact energy

Let $\xi=(x,y,z,0)$, $\chi=x-y+z$, and define the smooth factor

$$H_\nu(x,y,z)=\frac{e^{-\pi y}}{1024\pi^5}\prod_{j=1}^{4}\prod_{\varepsilon=\pm1}\Gamma\left(\frac14+\frac{i}{2}[\xi_j-\xi_{j+1}+\varepsilon\nu]\right),\quad \xi_5=\xi_1. \tag{12.6}$$

It is the F_Λ/Λ limit of (11.14), with its complex phase retained. In particular $4H_\nu(x,x+z,z)/\pi=v_\nu(x)v_\nu(z)$. Let $\mathcal{L}_T(x,y,z,0)$ denote the pullback of (12.4). Translation of all four labels does not change it. Then in the exact-energy variables of (9.7),

$$e^{it_M(x-y+z)}\widetilde{\mathcal{C}}_M(r;x,y,z)\longrightarrow H_\nu(x,y,z)\mathcal{L}_T(x,y,z,0)\quad\text{in }\mathcal{D}'(\mathbb{R}^4). \tag{12.7}$$

To verify that the changing time introduces no missing constant, change the four energy variables to their exact representation labels as in Theorem 11.2. The smooth test and external-factor errors are $O(M^{-1})$ on a compact set. The difference between the two linear phase arguments is at worst $O(M^{-1})$, so their exponentials differ by $O(M^{-1}\log M)$, uniformly for compact T. The pointwise compact bound $\Lambda S=O(1+\log\Lambda)$ bounds the resulting pairing error by $O(M^{-1}\log^2 M)$. The three internal Jacobians cancel in the change of variables; the external inverse Jacobian tends to one. For energy differences alone the sharper error is $O(M^{-2})$, but it is not needed. These estimates also justify replacing Λ by M in the logarithm.

12.3 Hard thresholds and arbitrary fixed packets

Put $Q=x^2+(y-x)^2+(y-z)^2+z^2$, and let K_0 be (9.9). The two limiting threshold weights after integrating the external energy are

$$W_F(x,y,z)=1,\qquad W_Q(x,y,z)=K_0\big(\max(0,-x,-y,-z)\big). \tag{12.8}$$

Define

$$\mathcal{X}_P(T)=\int_{\mathbb{R}^3}^{\mathrm{reg}}W_P(x,y,z)e^{-\sigma^2Q/2}H_\nu(x,y,z)\mathcal{L}_T(x,y,z,0)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z,\qquad P=F,Q. \tag{12.9}$$

The superscript specifies the following definition: use smooth compact external densities and path cutoffs, smooth every intermediate threshold, pair the resulting smooth test with the displayed distribution, and then remove these approximations with the bounds below. It does not silently multiply a general distribution by a discontinuous function. For a general L^1 density this definition is retained; no ordinary absolutely convergent three-dimensional integral or unsmoothed pointwise principal value is asserted.

Theorem 12.2 (Finite-packet limit and ordered plateaux)**.** *For every fixed probability density $\pi_0\in L^1(\mathbb{R}_+)$ and either protocol, the regularized expression (12.9) exists, is independent of the allowed approximations, and*

$$X_M^P(t_M,0,t_M,0)\longrightarrow\mathcal{X}_P(T) \tag{12.10}$$

uniformly on compact T intervals. It obeys

$$\lim_{T\to-\infty}\mathcal{X}_P(T)=X_\infty^P,\qquad \lim_{T\to+\infty}\mathcal{X}_P(T)=0, \tag{12.11}$$

where X_∞^P is the repeated-time fixed-time limit in Section 9. The limiting commutator norm squared is

$$\mathcal{D}_P(T)=2\big(N_\infty^P-\operatorname{Re}\mathcal{X}_P(T)\big). \tag{12.12}$$

Consequently $\mathcal{D}_F(-\infty)=0$, $\mathcal{D}_Q(-\infty)=\mathcal{D}_{\mathrm{edge}}$, and $\mathcal{D}_P(+\infty)=2N_\infty^P$.

Proof. All truncation estimates of Proposition 9.1 hold for every real time, so also for t_M. Each threshold layer of width η excludes a tree norm squared at most $2\eta A\sup B$, giving a pairing error $O(\sqrt{\eta})$. Smooth path cutoffs have the same positive tail control. Changing the external density in L^1 costs at most A^2 times its L^1 distance. These bounds are independent of M and T. For each fixed smooth approximation, (12.7) gives convergence. The approximation estimates pass to its limit and make the family of limiting pairings uniformly Cauchy for all T. This proves existence, approximation independence and compact-T uniform convergence. It also derives (12.8) from $r, r+x, r+y, r+z\ge 0$, rather than assuming the fixed-time delta constraint away from its support.

For a fixed compact smooth test, use the same Fourier identity on the minus branch of (12.4). Its singular part contributes $-2\operatorname{sgn}(T)\delta_D(\chi)/\pi$, while its smooth remainder vanishes. Thus $\mathcal{L}_T\to 4\delta_D(\chi)/\pi$ as $T\to-\infty$ and $\mathcal{L}_T\to 0$ as $T\to+\infty$. The approximation bounds just proved are uniform in T, allowing their removal after either limit. On $y=x+z$, the Gaussian, external factor and threshold weight recover exactly the fixed-time expression (9.9). The ordered norms are time independent and converge by the one-step positive-kernel limit and its Gaussian envelope. Equation (8.11), or direct expansion of the two-state norm, gives (12.12). □

Neither monotonicity nor an early-time exponential rate follows. In particular $\mathcal{D}_Q(T)-\mathcal{D}_{\mathrm{edge}}$ need not be positive from this argument. A hard moving boundary changes the analytic test class needed for contour shifts and cannot be ignored when extracting an exponent. The fixed-protocol limit is independent of π_0; the moving-protocol limit generally is not.

12.4 The constant time origin in the source

There is a useful direct coefficient check against KL (B.50), using the same raw ordering for this comparison. Let $a_\pm=q_1-q_2\pm\nu$ and define $R(z)=\Gamma(1-z)\sin(\pi z/2)$. The duplication and reflection identities give $c(z)=2^zR(z)$, so

$$V_+=2\pi\cosh(\pi\chi/2)\frac{R(p+v)R(q+v)}{R(p-t)R(q-t)}. \tag{12.13}$$

Here $p+v=1/2+ia_-$, $q+v=1/2+ia_+$, $p-t=1/2+i(a_--\chi)$ and $q-t=1/2+i(a_+-\chi)$. Reflecting the minus-ν Gamma factors in this expression gives exactly the coefficient multiplying $\Gamma(i\chi)$ in KL (B.50), after division by π^3. This is an algebraic identity off the pole, extended by meromorphy. For the source's negative phase convention, the exact assembled limit therefore has the surviving factor $e^{-iT\chi}\Gamma(i\chi)V_+/\pi^3$ at $\tau=\log(2\Lambda)+T$.

KL (6.32) uses that time origin, but (B.45) uses $\log(2\Lambda/e)$ with the same printed coefficient in (B.50). At one and the same physical time, write $T_{\mathrm{app}}=\tau-\log(2\Lambda/e)=T+1$. Our surviving factor is then $e^{-i(T_{\mathrm{app}}-1)\chi}$, whereas the printed (B.51) uses $e^{-iT_{\mathrm{app}}\chi}$. The difference is a finite $e^{i\chi}$ phase. Source footnote 73 acknowledges an additive shift; the calculation here fixes our convention from the exact Gamma ratios. Exact-energy corrections vanish and cannot supply this constant. We record this as a source-convention/asymptotic discrepancy, not an author-confirmed erratum. It does not affect the fixed-time delta or the two plateaux, but must be kept in any quantitative curve comparison.

13 Gaussian reduction and a hard-boundary tail

This section retains the model and the order of limits of Section 12. We now distinguish two early-time laws for the *limiting* correlator. The exponent two for an unprojected semiclassical OTOC is already present in KL; here we check the fixed Gaussian preparation explicitly. The hard moving threshold gives a different result for the specific density $\pi_0(r) = \beta e^{-\beta r}$, $\beta > 0$. No finite-M growing-window error estimate is supplied.

13.1 A factorization of the surviving branch

Change coordinates to

$$\chi = x - y + z, \quad u = x - \chi/2, \quad v = z - \chi/2, \qquad y = u + v, \tag{13.1}$$

with absolute Jacobian one. Define

$$f_\chi(w) = e^{-\pi w - \sigma^2 w^2} \prod_{\epsilon=\pm1} \prod_{\eta=\pm1} \Gamma\left(\frac{1}{4} + \frac{i\chi}{4} + \frac{i\eta}{2}(w + \epsilon\nu)\right),$$
$$J(\chi) = \int_{\mathbb{R}} f_\chi(w)\,\mathrm{d}w, \qquad K = \frac{J(0)}{16\pi^3} = \int_{\mathbb{R}} k(w)\,\mathrm{d}w. \tag{13.2}$$

For real χ, direct cancellation of the eight external Gamma factors against V_- gives

$$\frac{H_\nu V_-}{\pi^3} e^{-\sigma^2 Q/2} = \frac{2^{2i\chi}\cosh(\pi\chi/2)}{512\pi^7} e^{-\sigma^2\chi^2/2} f_\chi(u) f_\chi(v). \tag{13.3}$$

For example the x-dependent numerator of the c ratios in V_- is $c(1/2 - i(x+\nu))c(1/2 - i(x-\nu))$; its denominator replaces x by $x - \chi$. Inserting the defining Gamma ratios cancels precisely the oppositely oriented two external edges. The remaining x and z factors are equal after (13.1). Also $Q = 2u^2 + 2v^2 + \chi^2$. This verifies the phase $2^{2i\chi}$ and all normalizations without dropping external phases.

For the fixed threshold this gives the one-dimensional Fourier integral

$$\mathcal{X}_F(T) = \frac{K^2}{2} + \mathrm{PV}\int_{\mathbb{R}} e^{iT\chi} A_F(\chi)\,\mathrm{d}\chi,$$
$$A_F(\chi) = \frac{2^{2i\chi}}{512\pi^7}\Gamma(-i\chi)\cosh(\pi\chi/2) e^{-\sigma^2\chi^2/2} J(\chi)^2. \tag{13.4}$$

Only the scalar integral defining J remains inside the Fourier integral. Near zero $A_F = iK^2/(2\pi\chi) + O(1)$. Thus a useful ordinary-integral implementation is

$$\mathcal{X}_F(T) = \frac{K^2}{2}\left[1 - \mathrm{erf}(T/2)\right] + \int_{\mathbb{R}} e^{iT\chi}\left[A_F(\chi) - \frac{iK^2 e^{-\chi^2}}{2\pi\chi}\right]\mathrm{d}\chi. \tag{13.5}$$

The bracket has a removable value at zero. Real-axis conjugation gives $A_F(-\chi) = \overline{A_F(\chi)}$, hence $\mathcal{X}_F(T)$ is real.

13.2 Fixed threshold: the first surviving pole

Theorem 13.1 (Gaussian fixed-threshold early asymptotic)**.** *For fixed $\nu, \sigma > 0$ and any $2 < h < 4$,*

$$\mathcal{X}_F(T) = K^2 - C_{\nu,\sigma} e^{2T} + O_h(e^{hT}),$$
$$\mathcal{D}_F(T) = 2C_{\nu,\sigma} e^{2T} + O_h(e^{hT}), \qquad T \to -\infty,$$
$$C_{\nu,\sigma} = \frac{e^{2\sigma^2}}{32\pi^6} J(-2i)^2 > 0. \tag{13.6}$$

The constants are not asserted uniform as $\sigma \to 0$ or ∞.

Proof. In the closed lower strip $-h \le \operatorname{Im}\chi \le 0$, the real part of every Gamma argument in (13.2) is between $1/4$ and $1/4 + h/4$. Euler's integral bounds their moduli and all fixed parameter derivatives by constants uniform in the imaginary arguments. The w Gaussian therefore gives holomorphy of J and a uniform bound on each such strip. The Gamma asymptotic on vertical lines then bounds $\Gamma(-i\chi)\cosh(\pi\chi/2)$ by a polynomial in $|\operatorname{Re}\chi|$ away from its poles. The remaining Gaussian ensures horizontal integrability and vanishing vertical sides.

The odd poles cancel, as is manifest in the identity

$$\Gamma(-i\chi)\cosh(\pi\chi/2) = 2^{-i\chi-1}\sqrt{\pi}\,\frac{\Gamma(-i\chi/2)}{\Gamma((1+i\chi)/2)}. \tag{13.7}$$

Move the principal-value contour down to $\mathbb{R} - ih$. Clockwise orientation gives minus one half of $2\pi i$ times the zero-pole residue, and minus $2\pi i$ times each pole strictly below it. The zero pole adds $K^2/2$. The only other pole crossed is $\chi = -2i$, with

$$\operatorname*{Res}_{\chi=-2i} A_F = -\frac{ie^{2\sigma^2}J(-2i)^2}{64\pi^7}.$$

This yields the negative coefficient in (13.6). On the displaced line $|e^{iT\chi}| = e^{hT}$ and its remaining integral is absolutely bounded. Finally

$$J(-2i) = \int_{\mathbb{R}} e^{-\pi w - \sigma^2 w^2} \prod_{\epsilon=\pm1} \left|\Gamma\left(\frac{3}{4} + \frac{i}{2}(w + \epsilon\nu)\right)\right|^2 \mathrm{d}w > 0.$$

Since $N_\infty^F = K^2$, the commutator formula follows. □

This is an exponent for the ordered limit $M \to \infty$ followed by $T \to -\infty$ of the specified vacuum commutator norm. It is not, without a uniform finite-M remainder, a fitted Lyapunov exponent over an arbitrary experimental time window.

13.3 Moving threshold: an explicit corner contribution

For $\pi_0(r) = \beta e^{-\beta r}$, retain the hard projection at each insertion. Define

$$\begin{aligned} J_Q(\chi) &= \int_0^\infty \beta e^{-\beta r}\,\mathrm{d}r \iint_{D_{r,\chi}} f_\chi(u) f_\chi(v)\,\mathrm{d}u\,\mathrm{d}v, \\ D_{r,\chi} &= \{u, v \ge -r - \chi/2,\ u + v \ge -r\}. \end{aligned} \tag{13.8}$$

For real χ this is an ordinary convergent integral. Replacing J^2 by J_Q in A_F defines A_Q, and

$$\mathcal{X}_Q(T) = \frac{X_\infty^Q}{2} + \mathrm{PV}\int e^{iT\chi} A_Q(\chi)\,\mathrm{d}\chi, \qquad X_\infty^Q = \frac{J_Q(0)}{256\pi^6}. \tag{13.9}$$

The regularity proved next justifies this ordinary PV for the exponential density; it does not assert it for every L^1 density of Section 12.

Put $a = -r - \chi/2$, $L = r + \chi$ and

$$\begin{aligned} B(r,\chi) &= \int_{s,t\ge0,\ s+t<1} f_\chi(a + Ls) f_\chi(a + Lt)\,\mathrm{d}s\,\mathrm{d}t, \\ I(r,\chi) &= \left[\int_a^\infty f_\chi(w)\,\mathrm{d}w\right]^2 - L_+^2 B(r,\chi). \end{aligned} \tag{13.10}$$

Indeed the square of half-lines has a triangular forbidden region only when $L > 0$. If the factor L_+^2 is replaced by L^2, the resulting r integral is a smooth function $J_{\mathrm{sm}}(\chi)$ on the real line. The exact difference is

$$J_Q(\chi) - J_{\mathrm{sm}}(\chi) = \mathbf{1}_{\chi<0} \int_0^{-\chi} \beta e^{-\beta r}(r+\chi)^2 B(r,\chi)\,\mathrm{d}r = \frac{\beta f_0(0)^2}{6}(-\chi)_+^3 + O((-\chi)_+^4). \quad (13.11)$$

The Taylor remainder here also has the one-sided derivative bounds obtained by differentiating the rescaled simplex integral. In particular it is not merely a pointwise big-O assertion.

Theorem 13.2 (Hard-boundary inverse-cubic tail)**.** *For the stated exponential density, fixed $\beta, \nu, \sigma > 0$, define*

$$B_{\beta,\nu} = \frac{\beta f_0(0)^2}{1536\pi^7} = \frac{\beta|\Gamma(1/4 + i\nu/2)|^8}{1536\pi^7} > 0. \quad (13.12)$$

In the order of limits specified above,

$$\begin{aligned} \mathcal{X}_Q(T) &= X_\infty^Q + \frac{B_{\beta,\nu}}{T^3} + O(|T|^{-4}), && T \to -\infty, \\ \mathcal{X}_Q(T) &= \frac{B_{\beta,\nu}}{T^3} + O(T^{-4}), && T \to +\infty, \\ \mathcal{D}_Q(T) - \mathcal{D}_{\mathrm{edge}} &= \frac{2B_{\beta,\nu}}{|T|^3} + O(|T|^{-4}), && T \to -\infty. \end{aligned} \quad (13.13)$$

The error can be complex in the first two lines. The leading coefficient is independent of σ at fixed $\sigma > 0$; neither the error constant nor the range of useful asymptotic time is uniform in σ.

Proof. For real χ, w, all Gamma arguments in f_χ have real part $1/4$. Euler's integral, including its logarithmic moments, bounds all fixed derivatives uniformly in their imaginary parts. Derivatives in w introduce only polynomial factors from the Gaussian. Consequently J_{sm} and its derivatives grow at most polynomially on the real line: differentiating the square and the simplex introduces polynomial factors in r, χ, integrable against $\beta e^{-\beta r}$. The same statement holds on each half-line for the correction in (13.11). After the outer Gaussian, the amplitudes and all their one-sided derivatives decrease rapidly at infinity.

To obtain the corner coefficient set $\chi = -h$, $r = h\rho$ in the correction, $h > 0$. The measure and L^2 give h^3; the remaining integrand is smooth through $h = 0$. Its value integrates to $\beta f_0(0)^2 \int_0^1 (1-\rho)^2\,\mathrm{d}\rho/2 = \beta f_0(0)^2/6$, proving the stated differentiated expansion. Multiplying by $\Gamma(-i\chi) = i/\chi + O(1)$ gives the nonsmooth term in A_Q,

$$-ia_0\chi^2\mathbf{1}_{\chi<0}, \qquad a_0 = \frac{\beta f_0(0)^2}{3072\pi^7}, \quad (13.14)$$

up to higher one-sided powers and a smooth function near zero. Subtract also $iX_\infty^Q e^{-\chi^2}/(2\pi\chi)$ to remove the PV pole. The regular amplitude is C^1 and its second derivative has jump $2ia_0$ from left to right. Integration by parts on the two half-lines therefore gives $2a_0/T^3 + O(|T|^{-4})$. The rapidly decreasing one-sided fourth derivatives justify the remainder. The subtracted pole gives the two ordered plateaux, and $2a_0 = B_{\beta,\nu}$. Taking the real part in the commutator identity proves the final line. □

Thus even after subtracting the static floor, this hard-boundary protocol has an inverse-cubic early correction for the specified preparation. It cannot be assigned the fixed-protocol exponential law. The coefficient comes from $r = u = v = \chi = 0$, where the Gaussian is one. This is a preparation-boundary effect in the strict-$G_N \to 0$ model; it is not a new gravitational interaction. Other densities, especially those vanishing near $r = 0$, require their own endpoint analysis.

13.4 Reproducible diagnostics

For $\nu = 0.6$, $\sigma = 1$, the scalar integrals give approximately

$$K^2 = 0.02209171019693, \qquad C_{\nu,\sigma} = 0.01259281368446.$$

At $T = -8, -6, -4$, the empirical ratios $(K^2 - \mathcal{X}_F)/(C_{\nu,\sigma} e^{2T})$ are respectively 0.999981, 0.999020 and 0.969004. For $\beta = 1$ the moving-boundary commutator coefficient is $2B_{\beta,\nu} \simeq 0.00023770257895$. The archived code compares the real-axis subtracted integral with a lower contour, varies quadrature orders, and checks the local corner coefficient on a unit simplex. These are empirical numerical diagnostics, not interval certification or a finite-mass 6j computation. The full moving-protocol time curve has not been numerically resolved here.

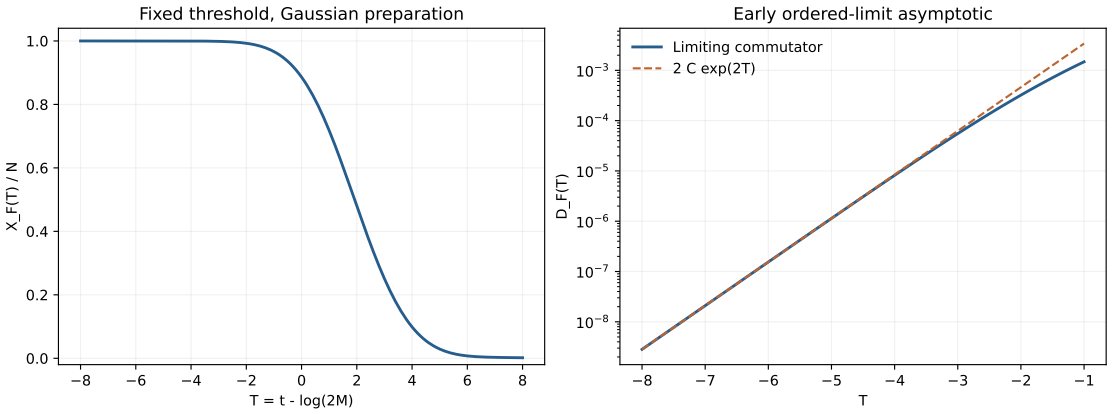


Figure 1: Fixed-threshold Gaussian limiting curve for $\nu = 0.6$, $\sigma = 1$. The right panel compares the commutator norm with its proven ordered-limit early asymptotic. Displayed curves have empirical quadrature checks; they are not finite-M data.

14 A resolved moving-threshold curve

We now evaluate the full limiting integral of Section 13, rather than constructing a curve from its local corner coefficient. The model remains dS_2, $L = 1$, proper $SO^+(2,1)$, strict $G_N \to 0$, BD free matter with two independent equal principal-series species, a normalized vacuum external packet, fixed proper Gaussian smearing, and the repeated moving projection Q_M. Here $\pi_0(r) = \beta e^{-\beta r}$. The numerical example fixes $\nu = 0.6$, $\sigma = \beta = 1$. All times use $t_M = \log(2M) + T$, with $M \to \infty$ taken first.

14.1 Splitting every transverse boundary

Integrating the external excess r exactly gives

$$J_Q(\chi) = \iint_{\mathbb{R}^2} f_\chi(u) f_\chi(v) \exp\left[-\beta \max\left(0, -u - \frac{\chi}{2}, -v - \frac{\chi}{2}, -u - v\right)\right] \mathrm{d}u\, \mathrm{d}v. \tag{14.1}$$

This is still the hard projection, with no smoothing of its boundaries. For fixed u, set

$$\begin{aligned} a_\chi(u) &= \max(0, -u - \chi/2), & b_\chi(u) &= \max(-\chi/2, -u), \\ E_\chi(s) &= \int_{-\infty}^{s} e^{\beta v} f_\chi(v)\, \mathrm{d}v, & H_\chi(s) &= \int_{s}^{\infty} f_\chi(v)\, \mathrm{d}v. \end{aligned} \tag{14.2}$$

The maximum in (14.1) is $\max(a_\chi, b_\chi - v)$. Its two branches meet at $v = b_\chi - a_\chi$, so

$$J_Q(\chi) = \int f_\chi(u) \left[e^{-\beta b_\chi} E_\chi(b_\chi - a_\chi) + e^{-\beta a_\chi} H_\chi(b_\chi - a_\chi) \right] \mathrm{d}u. \tag{14.3}$$

The outer panels split at $u = \pm\chi/2$, and each inner panel ends at the actual moving boundary. No tensor rule is applied across a max fold.

As an independent geometric representation, use symmetry to restrict to $v \le u$, and put $d_\chi(u) = \chi/2$ for $u \ge \chi/2$ and $d_\chi(u) = u$ otherwise. Then

$$J_Q(\chi) = 2 \int f_\chi(u) \left[e^{\beta d_\chi(u)} E_\chi(\min(u, -d_\chi(u))) + \mathbf{1}_{u > -d_\chi(u)} \int_{-d_\chi(u)}^{u} f_\chi(v)\, \mathrm{d}v \right] \mathrm{d}u. \tag{14.4}$$

Indeed, in this half-plane $-v - \chi/2 \ge -u - \chi/2$; comparison with $-u - v$ switches at $u = \chi/2$. Both implementations retain complex f_χ. They share the same special-function routine, so agreement checks geometry and quadrature, not independent Gamma certification.

14.2 Static normalization and stable small differences

At $\chi = 0$, the weight factorizes. With $k(w) = f_0(w)/(16\pi^3)$, define

$$\begin{aligned} g &= \int_0^\infty k(x)\, \mathrm{d}x, & \ell &= \int_0^\infty k(-u) e^{-\beta u}\, \mathrm{d}u, \\ m &= \int_0^\infty \int_0^\infty k(x) k(-u) e^{-\beta (u-x)_+}\, \mathrm{d}u\, \mathrm{d}x. \end{aligned} \tag{14.5}$$

The two-step path norm and early crossed plateau are

$$N_\infty^Q = g^2 + g\ell + m + \ell^2, \qquad X_\infty^Q = (g + \ell)^2, \qquad \mathcal{D}_{\mathrm{edge}} = 2(m - g\ell). \tag{14.6}$$

The last difference is computed directly as a positive integral with weight $e^{-\beta(u-x)_+}(1 - e^{-\beta \min(u,x)})$, using `expm1`. The resulting values are

$$\begin{aligned} N_\infty^Q &\simeq 0.007843474197608, \\ X_\infty^Q &\simeq 0.007649270098482, \\ \mathcal{D}_{\mathrm{edge}} &\simeq 0.0003884081982512. \end{aligned} \tag{14.7}$$

The independently evaluated $J_Q(0)/(256\pi^6)$ agrees with X_∞^Q.

For either subtraction width $a = 1$ or 2, let

$$I_a(T) = \int e^{iT\chi} \left[A_Q(\chi) - \frac{i X_\infty^Q e^{-a\chi^2}}{2\pi\chi} \right] \mathrm{d}\chi. \tag{14.8}$$

The exact representations used in the numerical calculation are

$$\begin{aligned} \mathcal{X}_Q(T) &= \frac{X_\infty^Q}{2} \left[1 - \operatorname{erf}\left(\frac{T}{2\sqrt{a}} \right) \right] + I_a(T), \\ \mathcal{D}_Q(T) - \mathcal{D}_{\mathrm{edge}} &= X_\infty^Q \left[1 + \operatorname{erf}\left(\frac{T}{2\sqrt{a}} \right) \right] - 2\,\mathrm{Re}\, I_a(T). \end{aligned} \tag{14.9}$$

The second line avoids subtracting two plateau-size commutators. It does not remove cancellation inside the Fourier integral. Unlike the fixed protocol, $\mathcal{X}_Q$ need not be real: for example $\mathcal{X}_Q(0) \simeq 0.00691324056 + 0.00008422460\, i$.

14.3 Numerical comparison with the corner prediction

Composite Gauss–Legendre integration uses a transverse box $[-10,10]^2$ and $\chi \in [-12,12]$, with a panel break at zero. Transverse/Fourier orders are varied separately as $(24,40)$, $(32,40)$ and $(24,64)$. The central Fourier panels have width $1/2$. Both PV subtraction widths, both transverse representations at selected frequencies, and transverse radii $8, 10, 12$ are recorded. Twenty-five detailed times and a 257-point central curve retain full complex values. The maximum changes in the 25 complex correlators are approximately 6.14×10^{-18} under transverse refinement and 2.08×10^{-17} under Fourier refinement. Relative to the real early plateau difference or the real late correlator, the largest observed change is 1.90×10^{-7}. These are empirical differences between rules, not rigorous accuracy estimates.

The actual integrals give dimensionless ratios to the leading coefficient of Theorem 13.2, defined by

$$R_{\mathrm{early}}(s) = \frac{(\mathcal{D}_Q(-s) - \mathcal{D}_{\mathrm{edge}})s^3}{2B}, \qquad R_{\mathrm{late}}(s) = \frac{\operatorname{Re}\mathcal{X}_Q(s)s^3}{B}.$$

$\lvert T\rvert$	$R_{\mathrm{early}}(\lvert T\rvert)$	$R_{\mathrm{late}}(\lvert T\rvert)$
16	0.74636	1.75824
32	0.85907	1.16537
64	0.92589	1.08068
128	0.96205	1.03962

These approach the predicted value one without a fitted time shift. Even at $|T| = 128$, finite-T corrections are several percent; the leading asymptotic is not an exact finite-time curve. At $T = -128$ the early difference is about 1.09044×10^{-10}, and the absolute regular-integrand scale divided by half that signal is about 7.28×10^7. Refinement comparisons must therefore be reported relative to this small signal, not only to N_∞^Q.

For real frequencies Euler's integral gives $|f_\chi(w)| \le \Gamma(1/4)^4 e^{-\pi w - w^2}$ at the stated $\sigma = 1$. If G is the integral of this envelope and g_R its tail outside $[-R,R]$, then the omitted transverse region contributes at most $2Gg_R$ to $J_Q(\chi)$, since the projection weight is at most one. The outer Gaussian similarly controls the omitted real-frequency tail. The archived bound expressions are evaluated in floating point, without outward rounding. They are separate tail bounds, not a certified total error for the subtracted PV quadrature. Order comparisons provide empirical numerical support for the curve and the corner prediction.

15 Preparation endpoints, phases, and the real commutator

We retain the state, free-matter algebra, projected Gaussian operators, proper group, units, and strict-$G_N \to 0$ assumptions of Section 14. We change only the normalized external excess density. The limit remains $M \to \infty$ at $t_M = \log(2M) + T$, followed by large positive or negative T. The endpoint Fourier expansion is standard (DLMF 2.3.4 [6]); the issue here is which complex coefficient survives taking the real part of this particular correlator. No priority claim is made.

15.1 Gamma densities and a two-term corner expansion

For a fixed integer $p \ge 0$ and $\beta > 0$, take

$$\pi_p(r) = \frac{\beta^{p+1}}{p!} r^p e^{-\beta r} \mathbf{1}_{r \ge 0}. \tag{15.1}$$

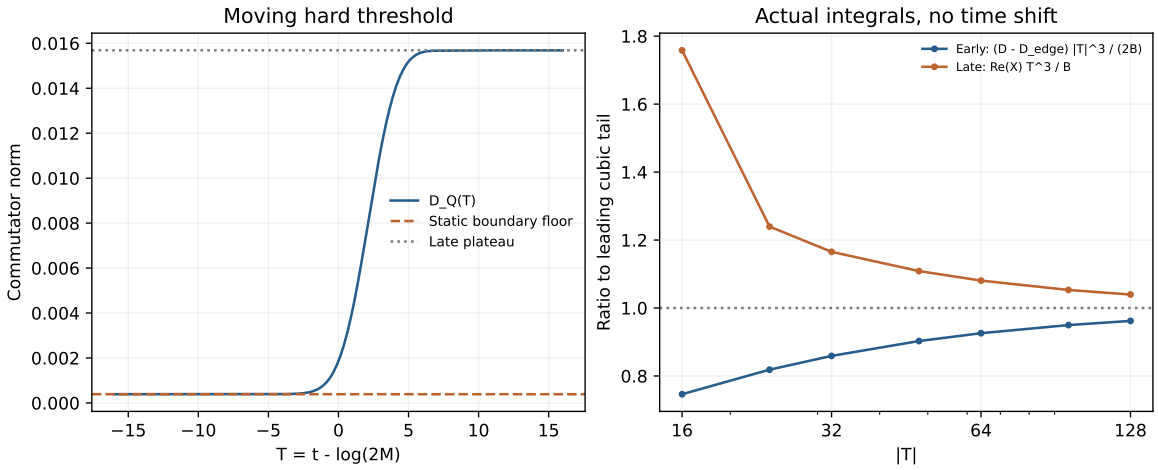


Figure 2: Actual limiting integrals for the exponential packet and repeated moving hard threshold. The right panel compares independently computed time curves with the analytic corner coefficient; the coefficient was not used to generate the curves. This is ordered-limit data, not a finite-mass simulation or finite-G_N calculation.

Write J_p for (13.8) with this density. The square-minus-triangle formula (13.10) still holds. Let $J_{p,\mathrm{sm}}$ denote the reference obtained by replacing L_+^2 by L^2 inside the r integral. Its exact difference is

$$J_p(\chi) - J_{p,\mathrm{sm}}(\chi) = \mathbf{1}_{\chi<0} \int_0^{-\chi} \pi_p(r)(r+\chi)^2 B(r,\chi)\,\mathrm{d}r. \tag{15.2}$$

This reference is an algebraic device, not a different projection. Define real constants

$$\begin{aligned} F_\nu &= f_0(0) = |\Gamma(1/4 + i\nu/2)|^4, \\ q_\nu &= \mathrm{Re}\,\psi(1/4 + i\nu/2), & H_\nu &= 2q_\nu + \gamma_E + 2\log 2, \\ d_p &= \frac{\pi p - \beta(p+1)}{p+4}, & S_p &= \frac{\beta^{p+1} F_\nu^2}{512\pi^7}. \end{aligned} \tag{15.3}$$

Here H_ν is a scalar coefficient, distinct from the external-leg function $H_\nu(x,y,z)$ of Section 12.

Lemma 15.1 (Differentiated endpoint expansion)**.** *For $h \downarrow 0$,*

$$J_p(-h) - J_{p,\mathrm{sm}}(-h) = \frac{\beta^{p+1} F_\nu^2}{(p+3)!} h^{p+3} \left[1 + h(d_p - 2iq_\nu) + O(h^2)\right]. \tag{15.4}$$

The bracket is smooth in h near zero, so the remainder has its corresponding one-sided derivative bounds.

Proof. Set $r = h\rho$, $\chi = -h$, $0 \le \rho \le 1$. In the unit triangle $s,t \ge 0$, $s+t<1$, its two arguments become

$$u = h\left(\frac{1}{2} - \rho + (\rho-1)s\right), \qquad v = h\left(\frac{1}{2} - \rho + (\rho-1)t\right).$$

At the origin the derivatives are

$$\partial_w \log f_\chi(w)\big|_{0,0} = -\pi, \qquad \partial_\chi \log f_\chi(w)\big|_{0,0} = iq_\nu. \tag{15.5}$$

The Gamma factors are even in w at zero, while their four digammas sum to $4q_\nu$. The Gaussian has no first derivative there. The triangle has area $1/2$ and mean $s+t = 2/3$, hence

$$B(h\rho, -h) = \frac{F_\nu^2}{2}\left[1 + h\left(-2iq_\nu - \frac{\pi}{3} + \frac{4\pi}{3}\rho\right) + O(h^2)\right].$$

Combining this with $e^{-\beta h\rho}$, use

$$\int_0^1 \rho^p(1-\rho)^2\,\mathrm{d}\rho = \frac{2p!}{(p+3)!}, \qquad \frac{\int_0^1 \rho^{p+1}(1-\rho)^2\,\mathrm{d}\rho}{\int_0^1 \rho^p(1-\rho)^2\,\mathrm{d}\rho} = \frac{p+1}{p+4}.$$

This gives (15.4). The rescaled integration domain is compact and all Gamma arguments stay away from poles in a common neighborhood of $h=0$. Differentiation under the integral proves the claimed remainder control. □

15.2 Taking the real part changes the leading power

Let $\mathcal{P}_p(T) = X^Q_{\infty,p}$ for $T<0$ and zero for $T>0$; this notation specifies the two asymptotic plateaux, not an additional time-domain term in the experiment.

Theorem 15.2 (Gamma-packet ordered-time tails)**.** *For fixed $p\in\mathbb{Z}_{\ge 0}$ and $\beta,\nu,\sigma>0$,*

$$\mathcal{X}_p(T) - \mathcal{P}_p(T) = S_p\left[\frac{-i}{(p+3)(iT)^{p+3}} + \frac{-H_\nu - id_p}{(iT)^{p+4}}\right] + O(|T|^{-p-5}), \qquad T\to\pm\infty. \tag{15.6}$$

Consequently, for $s\to+\infty$,

$$\begin{aligned}\mathcal{D}_p(-s) - \mathcal{D}_{\mathrm{edge},p} &= \frac{2(-1)^k S_p}{p+3}s^{-p-3} + O(s^{-p-4}), && p=2k,\\ &= 2(-1)^{k+1}S_p d_p\, s^{-p-4} + O(s^{-p-5}), && p=2k+1.\end{aligned} \tag{15.7}$$

If p is odd and $\beta=\pi p/(p+1)$, the displayed real coefficient vanishes; this theorem then supplies an $O(s^{-p-5})$ bound, not a nonzero leading coefficient at that order.

Proof. The prefactor multiplying J_p in the Fourier amplitude is

$$P_\sigma(\chi) = \frac{2^{2i\chi}}{512\pi^7}\Gamma(-i\chi)\cosh(\pi\chi/2)e^{-\sigma^2\chi^2/2}. \tag{15.8}$$

At $\chi=-h$ it equals

$$P_\sigma(-h) = \frac{1}{512\pi^7}\left[-\frac{i}{h} - \gamma_E - 2\log 2 + O(h)\right].$$

Thus the two nonsmooth amplitude terms, expressed in $h=-\chi>0$, are

$$\frac{S_p}{(p+3)!}\left[-ih^{p+2} + (-H_\nu - id_p)h^{p+3}\right]\mathbf{1}_{h>0}. \tag{15.9}$$

For a smooth cutoff equal to one at $h=0$, the endpoint transform of h^n under e^{-iTh} has leading term $n!/(iT)^{n+1}$. This follows by repeated integration by parts on the half-line, with the same formula for either sign of real T. It is not an absolutely convergent transform of an uncut polynomial.

For completeness, the needed global bounds follow from Euler's Gamma integral and its logarithmic moments. All fixed real derivatives of f_χ are bounded by a Gaussian in w times a polynomial in w. The square and rescaled-triangle terms and their derivatives have at most polynomial growth in r,χ; every r moment of (15.1) is finite. Multiplication by the outer Gaussian makes all one-sided amplitude derivatives integrable at infinity. Subtracting the PV pole $iX^Q_{\infty,p}e^{-\chi^2}/(2\pi\chi)$ removes the remaining zero singularity. A local smooth cutoff separates (15.9) from a smooth rapidly decreasing part. The differentiated Taylor remainder and one further integration by parts give $O(|T|^{-p-5})$. The subtracted pole supplies the two plateaux, up to Gaussian-small terms. This proves (15.6). Finally use $\mathcal{D}_p = 2(N^Q_{\infty,p} - \mathrm{Re}\,\mathcal{X}_p)$ and $\mathcal{D}_{\mathrm{edge},p} = 2(N^Q_{\infty,p} - X^Q_{\infty,p})$. The first coefficient in (15.6) is real for even p and purely imaginary for odd p, giving (15.7). □

For $p = 0$ the leading term recovers Theorem 13.2. The first two complex coefficients are independent of σ at fixed $\sigma > 0$, because its Gaussian begins contributing at second order at this corner. Remainders and useful time ranges are not uniform as σ varies.

For $\beta = 1$, $p = 1$ has a purely imaginary T^{-4} term but a real T^{-5} correction. Since $d_1 = (\pi - 2)/5 > 0$, its early $\mathcal{D}_1 - \mathcal{D}_{\mathrm{edge},1}$ is negative. The $p = 2$ early correction is also negative. Therefore $\mathcal{D}_{\mathrm{edge}}$ is an early plateau, not a universal lower bound on $\mathcal{D}(T)$. This is compatible with positivity of the full commutator norm: its plateau is strictly positive. It also precludes universal monotone growth from the early plateau across these preparations.

15.3 A support gap moves the singularity away from the PV pole

Now take a fixed gap $g > 0$ and

$$\pi_g(r) = \beta e^{-\beta(r-g)} \mathbf{1}_{r \geq g}. \tag{15.10}$$

The projection is still Q_M, not Q_{M+g}. Only the external state has changed. The triangle correction begins at $\chi = -g$. Writing $\chi = -g - h$, $r = g + h\rho$ gives

$$J_g(-g-h) - J_{g,\mathrm{sm}}(-g-h) = \frac{\beta f_{-g}(-g/2)^2}{6} h^3 + O(h^4).$$

This again has differentiated one-sided remainders. Now P_σ is regular and nonzero at $-g$, so no zero-frequency pole lowers the corner order.

Proposition 15.3 (Gapped exponential packet)**.** *Define the nonzero complex number*

$$C_g = \beta P_\sigma(-g) f_{-g}(-g/2)^2. \tag{15.11}$$

With the analogous early/late plateau notation,

$$\begin{aligned} \mathcal{X}_g(T) - \mathcal{P}_g(T) &= \frac{C_g e^{-igT}}{T^4} + O(|T|^{-5}), && T \to \pm\infty, \\ \mathcal{D}_g(-s) - \mathcal{D}_{\mathrm{edge},g} &= -\frac{2\,\mathrm{Re}(C_g e^{igs})}{s^4} + O(s^{-5}), && s \to +\infty. \end{aligned} \tag{15.12}$$

The constants are for fixed $g > 0$; the expansion is not uniform as $g \downarrow 0$.

Proof. Apply the same cutoff and derivative argument at $\chi = -g$. The transform of h^3 supplies $6/(iT)^4 = 6/T^4$, and translation supplies e^{-igT}. The smooth part at $\chi = 0$ retains its PV pole and therefore the usual plateaux. All external moments remain finite. Gamma has no zeros and no pole at the arguments in (15.11), so $C_g \neq 0$. □

The signed early correction thus has oscillatory sign changes along sequences of arbitrarily early ordered-limit times. Removing density near $r = 0$ has not restored the fixed-threshold exponential: it has displaced the nonsmooth endpoint. Taking $g \to 0$ before the time asymptotic requires a separate crossover analysis.

15.4 Direct integral checks

For (15.1), the survival weight is $e^{-\beta q} \sum_{j=0}^{p} (\beta q)^j / j!$ for $q \geq 0$ (DLMF 8.4.8). The gapped exponential uses $e^{-\beta(q-g)_+}$. In both cases q is the actual max in (14.1). The first implementation integrates polynomial moments over exact slice domains; the second uses the survival function directly on the symmetric half-plane. No endpoint coefficient is used in generating the full curves.

The archived runs vary transverse/Fourier orders as $(24, 40)$, $(32, 40)$, and $(24, 64)$ for $p = 1, 2$ at $\beta = 1$ and for $g = \beta = 1$. They keep $\nu = 0.6$, $\sigma = 1$, complex amplitudes, and separate small real differences. Local corner integrals separately test the two-term Taylor coefficient. Numerical diagnostics and their precision limits are reported with the data; they do not independently certify these proof drafts, priority, or a total quadrature error.

For the numerical parameters, $H_\nu \simeq -1.65594788424$ and the early real coefficients multiplying s^{-5} are approximately $-1.62815710723\times10^{-4}$ for $p = 1$ and $-1.42621547371\times10^{-4}$ for $p = 2$. The directly integrated real corrections divided by these signed leading terms are

$s = \lvert T\rvert$	$p = 1$	$p = 2$
16	0.65345	0.62126
32	0.78681	0.77729
64	0.8824	0.8800

At $T = -64$, the actual corrections are about -1.3380×10^{-13} and -1.1689×10^{-13}, respectively. The largest observed relative changes of the small real component over the detailed times with $|T| \le 64$ are 4.64×10^{-4} and 6.62×10^{-4}. At $|T| = 128$, the signals are only a few 10^{-15} and their observed changes rise to about 0.65% and 0.95%. Those samples are retained but excluded from the primary quantitative table. The visible difference from unity is a finite-time effect, much larger than the displayed empirical refinement differences.

For $g = 1$, $C_g \simeq 0.000479119472 + 0.000588336141\,i$. The normalized signed oscillation in Figure 3 has maximum observed refinement change about 1.74×10^{-5} over its 513 plotted times. Its remaining phase and amplitude corrections are not removed by fitting. The survival weights in all three examples are between zero and one, so the separate real-frequency and transverse envelope bounds of Section 14 continue to apply. They still do not certify the total PV quadrature error. The archive contains 384 diagnostics, full complex spectra, local-corner data, and the reduced-precision far-time samples.

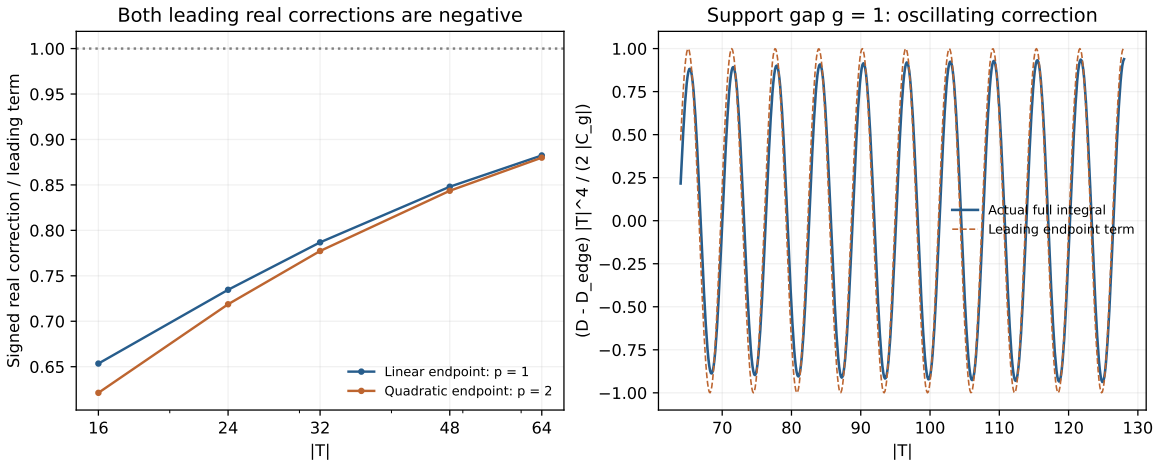


Figure 3: Actual full integrals versus endpoint predictions for $\nu = 0.6$, $\sigma = \beta = 1$. Left: ratios to the negative early s^{-5} corrections for Gamma packets. Right: the signed early correction for a fixed support gap $g = 1$, rescaled by $s^4/(2|C_g|)$, compared with $-\cos(\arg C_g + gs)$. These are empirical ordered-limit calculations, not finite-M data.

16 Keeping both branches at finite mass

We retain the dS_2, $L = 1$, proper-group, strict-$G_N \to 0$ model, BD vacuum, two equal independent principal-series species, fixed $\nu, \sigma > 0$, and the repeated projections of Section 9.

The external packet is normalized in exact energy. This section separates an identity for the assembled leading kernel from its error relative to the exact finite-mass correlator. All proof statements are author self-reviewed drafts. The two branches originate in KL (B.41)–(B.50) [1]; no new gravitational interaction or priority claim is intended.

16.1 Conjugate amplitudes with the actual projection

Put $\ell = \log(2M)$, $T = t - \ell$, and $k = t + \ell = 2\ell + T$. For either protocol let $A_P(\chi)$ denote the integrated minus-branch amplitude of Section 13, with the density replacements of Section 15 when appropriate. The plus-branch amplitude at the *same* real χ is $\overline{A_P(\chi)}$. This is not the claim $A_Q(-\chi) = \overline{A_Q(\chi)}$, which generally fails because the projection domain changes with χ. Indeed the pointwise identity, before any transverse integration, is

$$\frac{H_\nu V_+}{\pi^3} e^{-\sigma^2 \mathcal{Q}/2} = \frac{2^{-2i\chi}\cosh(\pi\chi/2)}{512\pi^7} e^{-\sigma^2\chi^2/2} f_{-\chi}(u) f_{-\chi}(v). \tag{16.1}$$

For example, the c-ratio numerator in V_+ contains $c(1/2 + i(x - y \pm \nu))$, and the denominator contains $c(1/2 - i(z \pm \nu))$. They cancel the corresponding oriented external Gamma factors. The four remaining x factors have common real part $1/4$ and shift $-i\chi/4$, and likewise for z. Since $f_{-\chi}(w) = \overline{f_\chi(w)}$ for real χ, w, comparison with (13.3) proves the identity. Multiplication by $\Gamma(i\chi)$ and integration against the same real density and region $D_{r,\chi}$ preserve conjugation.

Define the two-branch leading packet by

$$X_M^{(2),P}(t) = \mathrm{PV} \int_{\mathbb{R}} \left[e^{iT\chi} A_P(\chi) + e^{ik\chi} \overline{A_P(\chi)} \right] \mathrm{d}\chi. \tag{16.2}$$

The two pole residues cancel before integration; for the Gamma and gapped packets this is an ordinary integral of their combined amplitude. For other fixed L^1 densities use the common approximation prescription of Section 12. The superscript (2) identifies a leading-kernel quantity, not an exact 6j evaluation.

Proposition 16.1 (Mirror identity)**.** *With X_∞^P the real fixed-time crossed plateau,*

$$X_M^{(2),P}(t) = \mathcal{X}_P(t - \ell) + \overline{\mathcal{X}_P(-t - \ell)} - X_\infty^P, \tag{16.3}$$

$$\begin{aligned} D_M^{(2),P}(t) &:= 2\big(N_\infty^P - \mathrm{Re}\, X_M^{(2),P}(t)\big) \\ &= \mathcal{D}_P(t - \ell) + \mathcal{D}_P(-t - \ell) - \mathcal{D}_{\mathrm{edge},P}. \end{aligned} \tag{16.4}$$

Here $\mathcal{D}_{\mathrm{edge},F} = 0$. In particular $X_M^{(2),P}(-t) = \overline{X_M^{(2),P}(t)}$ and $D_M^{(2),P}$ is even.

Proof. Each Fourier integral in (16.2) is the corresponding $\mathcal{X}$ minus $X_\infty/2$. Complex conjugation changes the sign of the time argument. Adding them proves (16.3), and taking real parts proves (16.4). The same calculation works for every real smooth compact cutoff before its removal, so also for the regularized definition. □

The fast-branch remainder relative to its delta limit is therefore $\overline{\mathcal{X}_P(-k) - X_\infty^P}$, with no fitted time shift. It retains the full preparation-dependent early tail.

16.2 A compact estimate that does not differentiate the time phase

Let $0 \le r \le L$ and $|x|, |y|, |z| \le R$, with L, R fixed. Denote this box by $\Omega_{L,R}$. Set

$$C_M^0(x, y, z) = \frac{H_\nu(x, y, z)}{\pi^3} \left[e^{i\ell\chi} \Gamma(i\chi) V_+ + e^{-i\ell\chi} \Gamma(-i\chi) V_- \right], \quad \chi = x - y + z. \tag{16.5}$$

The bracket is evaluated at its removable value when $\chi = 0$.

Lemma 16.2 (Uniform compact density error)**.** *For $0 < \epsilon < 1/4$ and sufficiently large M, there is a constant $C_{L,R,\epsilon}$ independent of M and of all real t such that*

$$\sup_{\Omega_{L,R}} |\widetilde{\mathcal{C}}_M - C_M^0| \le C_{L,R,\epsilon}\, e_\epsilon(M),$$
$$e_\epsilon(M) = M^{-1+\epsilon} + \frac{1+\ell}{M} + \frac{(1+\ell)^2}{M^2}. \tag{16.6}$$

The same bound times the finite weighted volume controls the truncated crossed integral, uniformly over all real physical times t.

Proof. Use $\Lambda = M$ in Theorem 11.1. On the box,

$$s(M+\xi) - M = \xi - \frac{1}{8M} + O_{L,R}(M^{-2}), \qquad j(M+\xi) = 1 + O_{L,R}(M^{-2}).$$

The common $-1/(8M)$ cancels in every label difference. The external factor divided by M differs from H_ν by $O_{L,R}(M^{-1})$. The assembled bracket is $O_{L,R}(1+\ell)$ and its first offset derivatives are $O_{L,R}((1+\ell)^2)$. To see this through $\chi = 0$, subtract its explicit sinc term (11.11); its smooth remaining amplitudes multiply $e^{\pm i\ell\chi}$. Differentiating the sinc costs at most $O(\ell^2)$. This controls the $O(M^{-2})$ change of differences without differentiating the unknown exact remainder. Multiplication by the external factor and the three Jacobians gives (16.6). Crucially, both densities are compared in the *same exact energy coordinates*, then multiplied by the identical unit-modulus factor $e^{it\chi}$. There is no $|t|$ loss in this comparison. Bounded hard indicators can be retained in its ordinary compact integral. □

This improves the deliberately coarse phase-change argument in Section 12. It does not supply bounds on $C_{L,R,\epsilon}$ as the box grows with M.

16.3 Uniform absolute approximation on the full time axis

Let A, B, b be the positive-envelope quantities in Section 9 and put

$$\tau_X(L,R) = 2A\sqrt{2Ab(R/2)} + A^2 \int_L^\infty \pi_0(r)\,\mathrm{d}r,$$
$$\tau_N(L,R) = 2Ab(R/2) + A^2 \int_L^\infty \pi_0(r)\,\mathrm{d}r. \tag{16.7}$$

The external tail is a Gamma survival function or a shifted exponential for the packets of Section 15, and b is a Gaussian tail. Their constants remain those of the analytic envelope, not certified numbers.

Theorem 16.3 (Two-branch absolute approximation)**.** *For either fixed protocol and every fixed normalized L^1 external density, with fixed $\nu, \sigma > 0$,*

$$\sup_{t\in\mathbb{R}} |X_M^P(t,0,t,0) - X_M^{(2),P}(t)| \longrightarrow 0. \tag{16.8}$$

More precisely, for each fixed box and sufficiently large M,

$$\sup_t |X_M^P - X_M^{(2),P}| \le C'_{L,R,\epsilon} e_\epsilon(M) + 4\tau_X(L,R),$$
$$|N_M^P - N_\infty^P| \le C''_{L,R}/M + 2\tau_N(L,R). \tag{16.9}$$

Consequently $\delta_M^P := \sup_t |D_M^P(t) - D_M^{(2),P}(t)| \to 0$. No explicit rate for this supremum is claimed.

Proof. Truncate each ordered tree as in Proposition 9.1 and discard $r > L$. The exact crossed error is at most τ_X, uniformly in M, t. The truncated two-branch expression satisfies the same mirror identity, with truncated limiting curves $\mathcal{X}_{L,R}$ and plateau $X_{\infty,L,R}$. These curves can first be defined with smooth cutoffs and thresholds; the norm bounds allow their removal. Equivalently, on this compact box the transverse integral has a locally Lipschitz amplitude at $\chi = 0$: moving any of the finitely many linear faces changes a volume by $O(|\Delta\chi|)$, uniformly in bounded r, and the remaining smooth integrand has bounded derivatives. Integration against the L^1 density preserves this bound. Its pole-subtracted PV thus exists, including the hard faces.

For each fixed argument U, take the logarithmic-time limit of the exact full and truncated pairings. Their difference is at most τ_X, so $|\mathcal{X}_P(U) - \mathcal{X}_{L,R}(U)| \le \tau_X$ for *every* real U. This gives a supremum bound without interchanging a supremum and a convergence limit. The fixed-time limit likewise gives $|X^P_\infty - X_{\infty,L,R}| \le \tau_X$. The mirror identity now bounds the full-minus-truncated two-branch expression by $3\tau_X$, independently of M, t. On the compact box Lemma 16.2 bounds the remaining exact-minus-leading difference. This proves the first inequality in (16.9).

The positive ordered norms have excluded mass at most τ_N. Their compact one-step Gamma-ratio error is $O_{L,R}(M^{-1})$; passing the same tail bound to the limit proves the second inequality. For the fixed threshold the boundary has left the box once M is large enough. The moving threshold is identical in both integrands. Finally, first choose L, R large at fixed parameters, then take M large. The commutator error is at most twice the crossed error plus twice the norm error, proving the assertion. □

This theorem keeps both branches. Replacing the second branch by its delta before applying the theorem would give a different approximation and would discard the correction examined next.

16.4 Which early signals can the fast branch compete with?

Write $t = \ell - s$ so that $T = -s$ and $k = 2\ell - s$. Assume $s, k \to \infty$. Let $E_P(s) = \mathcal{D}_P(-s) - \mathcal{D}_{\mathrm{edge},P}$. Exactly,

$$D^{(2),P}_M(\ell - s) - \mathcal{D}_{\mathrm{edge},P} = E_P(s) + E_P(k). \tag{16.10}$$

For a Gamma packet with nonzero real coefficient c_p in (15.7), put $n = p + 3$ for even p and $n = p + 4$ for odd p. Then

$$\begin{aligned} D^{(2),p}_M(\ell - s) - \mathcal{D}_{\mathrm{edge},p} &= c_p(s^{-n} + k^{-n}) + O(s^{-n-1} + k^{-n-1}), \\ \frac{E_p(k)}{E_p(s)} &= \left(\frac{s}{k}\right)^n [1 + O(s^{-1} + k^{-1})]. \end{aligned} \tag{16.11}$$

In particular, for $s = a\ell$ with fixed $0 < a < 2$, the relative fast-branch contribution tends to $(a/(2-a))^n$, not zero. For positive physical times in this scaling take $0 < a < 1$. Dropping the fast branch is relatively valid for these power tails only when $s/k \to 0$, equivalently $s = o(\ell)$ in the early regime. The critical odd-p density with $c_p = 0$ is excluded. At fixed T, the fast residual itself has a power of $\log M$; this is a statement about the two-branch expression, not a determined leading rate for the exact-minus-single-branch correlator.

For the fixed-threshold Gaussian, (13.6) instead gives, for $2 < h < 4$,

$$D^{(2),F}_M(\ell - s) = 2C_{\nu,\sigma}(e^{-2s} + e^{-2k}) + O(e^{-hs} + e^{-hk}). \tag{16.12}$$

Its fast/slow ratio is $e^{-4t}[1 + o(1)]$, so relative suppression requires $t \to +\infty$ while $s \to \infty$. For a fixed gap,

$$D^{(2),g}_M(\ell - s) - \mathcal{D}_{\mathrm{edge},g} = -2\,\mathrm{Re}\left[C_g\left(\frac{e^{igs}}{s^4} + \frac{e^{igk}}{k^4}\right)\right] + O(s^{-5} + k^{-5}). \tag{16.13}$$

Near zeros of the first oscillatory term only an envelope comparison is meaningful; there is no uniform relative estimate there.

To transfer any displayed small signal to the exact finite-M observable one must additionally have $\delta_M^P = o(s^{-n})$, $o(e^{-2s})$, or $o(s^{-4})$, respectively. Theorem 16.3 proves $\delta_M^P \to 0$, not these explicit rates. It does imply existence of some slowly growing windows. For a noncritical Gamma packet one may take, for sufficiently large M,

$$s(M) = \min\{\sqrt{\ell}, (\delta_M^P)^{-1/(2n)}\},$$

with the second entry interpreted as infinity at zero error. Then $s \to \infty$, $s = o(\ell)$ and $\delta_M^P s^n \to 0$. For F replace the second entry by $\frac{1}{4}\log(1/\delta_M^F)$, which gives $\delta_M^F e^{2s} \to 0$. These are existence constructions in terms of an unknown error, not computable experimental windows. Quantifying the growth of the compact constants in (16.9) is still required.

16.5 Checks and the remaining finite-mass task

The accompanying calculation verifies (16.1) directly from the original c ratios at high precision. It then reuses the frozen full projected spectra of Sections 14–15, combines the two amplitudes before quadrature, and compares with (16.3). Small early differences are formed from the pole-subtracted integrals, not by subtracting two plateau-sized values. The numerical records distinguish exact algebra, empirical quadrature, and the still-uncomputed exact finite-mass remainder.

At $\nu = 0.6$, $\sigma = \beta = 1$, 300 empirical diagnostics pass. For $s = \ell/2$ and $\ell = 40$, the directly integrated fast/slow real-correction ratios for $p = 0, 1, 2$ are respectively 0.043283, 0.005156, and 0.005305. Their asymptotic targets are $1/27$, $1/243$, and $1/243$. The three independent stored quadrature rules give maximum relative changes of the small real quantities of 4.14×10^{-8}, 4.30×10^{-4}, and 6.69×10^{-4}, respectively; the gapped case gives 3.60×10^{-6}. These are empirical refinement differences, not certified total errors. The $p = 2$ slow correction has not reached its negative asymptotic regime at $\ell = 16$; its negative fast/slow ratio there is retained in the data. Figure 4 shows only $\ell \geq 24$ to resolve the subsequent approach, without treating earlier sign changes as numerical failures.

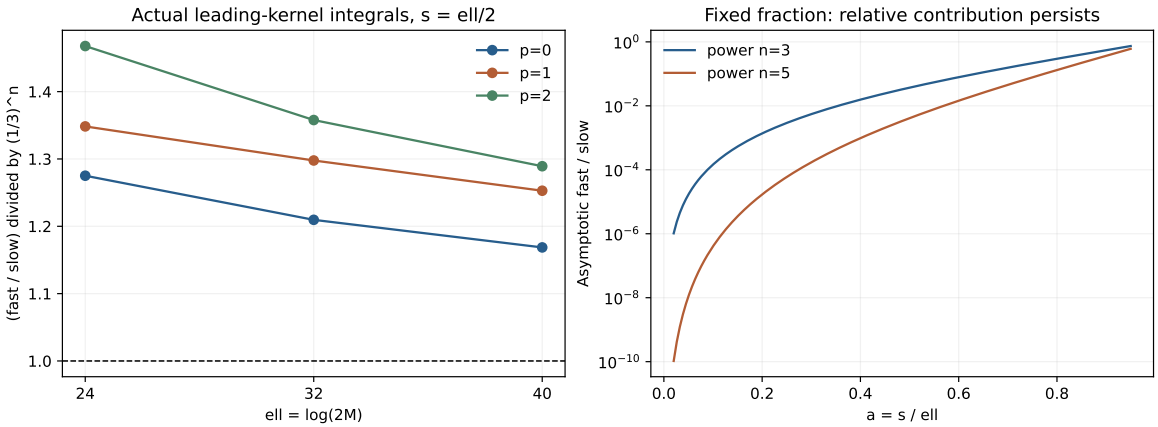


Figure 4: Left: actual two-branch leading-kernel packet integrals, with $s = \ell/2$ and $k = 3\ell/2$, normalized by the asymptotic fast/slow ratio. Right: the analytic limiting ratio for fixed $a = s/\ell$. Neither panel is an exact finite-M 6j calculation.

The next task is to bound $C'_{L,R,\epsilon}$ and $C''_{L,R}$ for an explicit growing choice of L, R. Gaussian path tails suggest R of order $\sqrt{\log\log M}$ and exponential external tails suggest L of order $\log\log M$ for inverse-logarithmic target errors, but these choices are not justified until the

local constants are tracked. No threshold smoothing loss is necessary for the compact two-branch comparison; the positive-tail argument controls its removal globally. Exact 6j packet calculations and total certified numerical errors remain separate open work, as do finite-G_N dynamics and a complete dual.

17 Growing cutoffs and explicit asymptotic time windows

We keep the states, projected Gaussian operators, exact-energy convention, proper group and strict-$G_N \to 0$ assumptions of Section 16. The issue is the dependence of its compact constants on the offsets. We give deliberately coarse analytic bounds, sufficient for explicit asymptotic windows. They do not provide a certified numerical minimum mass for a prescribed accuracy. The proofs below are author self-reviewed drafts, using the Euler and Stirling estimates of DLMF [6, 5.11, 15.6, 15.8, 2.3] and the exact KL kernel [1]; no priority claim is made.

17.1 Constants in the Euler endpoint argument

Let $Q_0 \geq 1$ bound the absolute values of the four real representation offsets, with fixed $\nu > 0$. The symbol Q_0 is unrelated to either projection Q_M or the Gaussian quadratic form $\mathcal{Q}$. All the small parameters in Sections 10–11 then have imaginary parts bounded by $c_0(1 + Q_0)$, where c_0 is independent of M, Q_0. Fix $0 < \epsilon < 1/4$. A complex tube of radius $\rho_\epsilon > 0$ can be chosen independently of Q_0 so that every Euler endpoint real part is between $1/4$ and $3/4$, the Gauss denominator real parts stay near one, and $|\operatorname{Re}\delta| < \epsilon/4$. The balanced identity remains exact. We take smaller nested disks when removing the apparent $e = 0$ and $\delta = 0$ poles.

We will use an envelope

$$\mathcal{K}(Q_0) = Ce^{cQ_0}(1 + Q_0)^d. \tag{17.1}$$

The fixed constants may be enlarged a finite number of times below; they depend on ν, ϵ and the tube margins, not on Q_0 or M. Products and finite sums of such envelopes have the same form.

Lemma 17.1 (Parameter-dependent endpoint bounds)**.** *For the four KL hypergeometric factors in Section 11, the endpoint remainder in (10.4), away from the small pole disks, is bounded by*

$$\mathcal{K}(Q_0)M^{-1+\epsilon/4}, \qquad M \geq C_0(1 + Q_0)^2. \tag{17.2}$$

The same envelope class controls the required finite derivative norms of the endpoint amplitudes. The constants and the large-mass threshold are uniform over the indicated growing offset boxes.

Proof. There are four separate estimates, with fixed differentiation order.

First, Gamma functions and their reciprocals on fixed vertical strips, away from the finitely many poles in those strips, are bounded by an exponential in the imaginary part times a polynomial. Stirling's formula on the tails and compactness on the bounded part give this bound; fixed derivatives obey it as well. Thus every small-parameter Gamma product, connection coefficient $G_\pm$, and c ratio off the pole disks is bounded by (17.1). The pole distance is fixed by ρ_ϵ, not by Q_0.

Second, for $Z = \pm i\lambda$, λ proportional to M, and shifts $|a|, |b| \leq c_1(1 + Q_0)$ with bounded real parts, the sectorial Stirling remainder gives

$$\begin{aligned} &\frac{\Gamma(Z + a)}{\Gamma(Z + b)} Z^{b-a} = e^{E(Z;a,b)}, \\ &\quad |E(Z; a, b)| \leq C_1(1 + Q_0)^2/M, \qquad M \geq C_0(1 + Q_0)^2. \end{aligned} \tag{17.3}$$

Here E is defined using the analytic log-Gamma branch in the fixed sector about $\pm i\mathbb{R}_+$, rather than the principal logarithm of the final ratio. To see the bound directly, write

$$E = \int_b^a [\psi(Z+z) - \log Z]\,\mathrm{d}z.$$

On the joining segment $|Z+z| \asymp M$ and its argument stays away from the negative real axis. The digamma expansion with sectorial remainder bounds its integrand by $C(1+Q_0)/M$; the segment length is $O(1+Q_0)$. Exponentiating gives a relative error bounded by $C(1+Q_0)^2/M$ after increasing C_0. This derives a growing-shift estimate, rather than substituting growing parameters into a fixed-shift ratio formula. Factors such as Z^{iy} cost at most e^{CQ_0}.

Third, all Gauss functions needed near either endpoint have an Euler integral with real positive bases. For $0 \le z \le z_* < 1$, $\operatorname{Re} B, \operatorname{Re}(C-B) > 0$, and fixed $j \le 5$, the ordinary Gauss function obeys

$$|\partial_z^j {}_2F_1(A,B;C;z)| \le |(A)_j| \left|\frac{\Gamma(C)}{\Gamma(B)\Gamma(C-B)}\right| \mathrm{B}(\operatorname{Re} B + j, \operatorname{Re}(C-B)) \times (1-z_*)^{-\max(\operatorname{Re} A + j, 0)}. \tag{17.4}$$

This follows by differentiating Euler's integral: $t, 1-t, 1-zt$ are positive, so their imaginary exponents have modulus one. It uses the ordinary function, with the $\Gamma(C)$ restored from DLMF 15.6.1. On the tube, the needed $(A,B;C)$ are $(b,d;g)$, $(b,d;1-\delta)$ and $(g-b,g-d;1+\delta)$; all satisfy the stated real-part conditions with fixed margins. Their derivative bounds have form (17.1). Near $w=0$ apply the latter two to $z = 1-e^{-w}$ on $0 \le w \le 1$; for w away from zero apply the first to $z = e^{-w}$.

Fourth, in (10.7), $h(z) = (1-e^{-z})/z$ is positive on the real half-line, $h(z) \asymp (1+z)^{-1}$, and its logarithmic derivatives of any fixed positive order are bounded there. Direct differentiation of J therefore gives, for $j \le 5$,

$$|J^{(j)}(w)| \le C_j (1+Q_0)^{d_j} (1+w)^{d_j} e^{-w/4}. \tag{17.5}$$

The t weights remain integrable with fixed endpoint margins. For $w \le 1$, the same bounds apply to the smooth factors multiplying w^{a-1} and w^0 in the connection expansion, with $a = 1-\delta$. Their first Taylor remainders have respective forms $w^a R_1(w)$ and $wR_2(w)$, with derivatives through order four bounded by $\mathcal{K}(Q_0)$. Choose a smooth cutoff at zero once and for all, independent of the parameters. Splitting at $w = 1/|Z|$ and integrating by parts four times on dyadic intervals gives transform bounds $\mathcal{K}(Q_0)|Z|^{-\operatorname{Re} a - 1}$ and $\mathcal{K}(Q_0)|Z|^{-2}$. Differentiating w^a only introduces fixed-degree polynomials in a; the dyadic sums are uniform because $\operatorname{Re} a$ stays near one. The exponentially decreasing part away from zero has four integrable derivatives, with the same envelope. The leading transforms are the Abel boundary values $\Gamma(a)Z^{-a}$ and Z^{-1}, with their cutoff errors handled by the same integrations by parts.

Finally, (17.3) controls the normalization $\mathcal{N}_M \mathrm{B}(\alpha,\beta)$ in Section 10. Multiplying the transform estimates by it, including its relative error, gives (17.2). No derivative order grows with Q_0. □

17.2 Assembly and the growth of the compact constants

Proposition 17.2 (Growing-box density control)**.** *There are $C, c, d, C_0 > 0$, for each fixed ν, σ, ϵ, such that the compact constants in (16.9) may be chosen as*

$$C'_{L,R,\epsilon} + C''_{L,R} \le C e^{c(L+R)} (1+L+R)^d, \qquad M \ge C_0 (1+L+R)^2. \tag{17.6}$$

The estimate retains the hard moving threshold. For F one also requires $M - R > m_*$, which holds eventually for the choices below.

Proof. For representation offsets bounded by Q_0, the exact beta-like prefactors have relative Gamma-ratio error $\mathcal{K}(Q_0)/M$, and size $\mathcal{K}(Q_0)M^{-1+\epsilon/4}$ off the apparent poles. Lemma 17.1, the exact cancellation (11.6), and the leading coefficients (11.7) consequently give

$$|S - S_M^0| \le \mathcal{K}(Q_0)M^{-2+\epsilon} \tag{17.7}$$

there, leaving room in ϵ for the fixed tube margins. The exact triple-integral holomorphy argument of Section 11 uses real parts of exponents only. Those margins are independent of the real offsets; hence its complex tube need not shrink with Q_0. Use fixed-radius disks in e, δ at their intersections. Their boundary stays in a box enlarged by a constant, on which the envelope has the same form. Maximum modulus transfers (17.7) inside. A pointwise choice of these disks has uniformly bounded overlap; one does not multiply the bound by an uncontrolled covering constant. Other Gamma arguments stay near real part $1/2$, so no new poles are encountered on enlarging the real offsets.

The eight large external Gamma factors satisfy the same growing-shift Stirling estimates. Their exponentially large mass factors cancel the hyperbolic prefactor exactly at leading order, as in (11.14). The remaining small Gamma factors and fixed offset derivatives have envelope (17.1). Thus $F_M/M - H_\nu$ is bounded by $\mathcal{K}(Q_0)/M$; the same type of bound applies to the one-step positive-kernel error. Hyperbolic errors are exponentially small for $M \ge C_0(1+Q_0)^2$ and are absorbed in these estimates.

In exact-energy coordinates $|\xi| \le L + R$, Taylor expansion gives

$$s(M+\xi) = M + \xi - \frac{1}{8M} + O((1+L+R)M^{-2}).$$

The Jacobians have uniform $O(M^{-2})$ error when $M > 2(L+R+1)$. After removal of the pole, the first derivatives of the leading bracket are bounded by $\mathcal{K}(Q_0)(1+\log(2M))^2$; this follows either by its explicit sinc subtraction or the small-Gamma bounds above. The argument of Lemma 16.2 now proves its density bound with an envelope of the stated form, taking $Q_0 = 1 + L + R$ and enlarging the constants for fixed ν. Compact volume factors add only a polynomial. The external density contributes its mass, at most one, rather than its supremum or derivatives. The Gaussian has modulus at most one. For the ordered norms, apply the same positive one-step estimate twice. No growing time seminorm or threshold smoothing constant is introduced. □

17.3 An explicit cutoff choice for exponential-moment packets

We now assume a fixed positive exponential moment:

$$B_0 := \int_0^\infty e^{\beta_0 r}\pi_0(r)\,\mathrm{d}r < \infty, \qquad \beta_0 > 0. \tag{17.8}$$

All fixed Gamma and shifted-exponential packets used above satisfy this, for example with $\beta_0 = \beta/2$; their B_0 are respectively 2^{p+1} and $2e^{\beta g/2}$. This extra assumption is not made for the arbitrary L^1 packets of Theorem 16.3.

Theorem 17.3 (Rate and asymptotic cutoffs)**.** *For a packet satisfying (17.8), both protocols obey, for every fixed $q > 0$,*

$$\delta_M^P := \sup_{t\in\mathbb{R}} |D_M^P(t) - D_M^{(2),P}(t)| = O_q(\ell^{-q}), \qquad \ell = \log(2M). \tag{17.9}$$

The same statement holds separately for the crossed error and the norm error. An explicit admissible cutoff choice, for sufficiently large M, is

$$L_M = \frac{q+2}{\beta_0}\log\ell, \qquad R_M = \sqrt{\frac{8(q+2)}{\sigma^2}\log\ell}. \tag{17.10}$$

There is also a positive mass-power rate $\delta_M^P = O(M^{-\kappa})$ for some $\kappa > 0$ depending on the envelope constants and the packet. We do not evaluate a numerical value of κ here.

Proof. Markov's inequality gives $\Pr(r > L) \le B_0 e^{-\beta_0 L}$. For $R \ge 1$, the Gaussian envelope of Section 9 gives

$$b(R/2) \le C e^{-\sigma^2 R^2/4}, \quad \tau_X + \tau_N \le C\{e^{-\sigma^2 R^2/8} + e^{-\beta_0 L}\}.$$

With (17.10), both tails are $O(\ell^{-q-2})$. Write $Q_0 = 1 + L_M + R_M = O_q(\log \ell)$. The compact term in Theorem 16.3 and Proposition 17.2 is then at most

$$C_q \ell^{a_q + o(1)} \left[M^{-1+\epsilon} + \frac{1+\ell}{M} + \frac{(1+\ell)^2}{M^2} \right],$$

for a finite constant a_q. For each fixed q this is smaller than every inverse power of ℓ, and $M \ge C_0 Q_0^2$ holds eventually. This proves (17.9), with no exchange of a fixed-box limit and a growing box.

For the last assertion set $\epsilon = 1/8$, let c be the enlarged coefficient in (17.6), and put

$$\kappa = \frac{\beta_0}{8(c+\beta_0)}, \quad L_M = \frac{\log M}{4(c+\beta_0)}, \quad R_M = \sqrt{\frac{16\kappa}{\sigma^2} \log M}.$$

Both tails are $O(M^{-2\kappa})$. The compact term is $M^{-7/8 + c/[4(c+\beta_0)] + o(1)}$, which is $O(M^{-\kappa})$ since $\kappa \le 1/8$. The other terms are smaller; the required large-mass condition still holds. The constants c, C_0 have analytic existence bounds above but have not been numerically certified. □

17.4 Consequences for the exact finite-mass observable

Let $c_p \neq 0$ be the real coefficient in (15.7), and $n = p + 3$ for even p, $n = p + 4$ for odd p. All density, field, Gaussian and projection parameters are fixed in the following limits. For every fixed $0 < \alpha < 1$, the *exact* commutator satisfies

$$D_M^p(\ell - \ell^\alpha) - \mathcal{D}_{\mathrm{edge},p} = c_p \ell^{-\alpha n} \left[1 + O\left(\ell^{-\alpha} + \ell^{-n(1-\alpha)} \right) \right]. \tag{17.11}$$

In particular $s = \sqrt{\ell}$, which was not justified by Section 16 alone, is now an explicit asymptotic window. The signed coefficient remains negative for $p = 1, 2$, $\beta = 1$. For a fixed fraction $0 < a < 2$ one instead obtains

$$D_M^p((1-a)\ell) - \mathcal{D}_{\mathrm{edge},p} = c_p \ell^{-n} \{ a^{-n} + (2-a)^{-n} \} + O(\ell^{-n-1}). \tag{17.12}$$

This estimate is uniform on closed subintervals of $0 < a < 2$. It retains both tails, including for positive physical time $a < 1$. For fixed physical t in a bounded interval, it implies

$$D_M^p(t) - \mathcal{D}_{\mathrm{edge},p} = 2 c_p \ell^{-n} + O(\ell^{-n-1}). \tag{17.13}$$

For bounded T at the positive logarithmic-time transition, it also gives

$$D_M^p(\ell + T) - \mathcal{D}_p(T) = c_p (2\ell + T)^{-n} + O(\ell^{-n-1}). \tag{17.14}$$

The critical odd-p case with $c_p = 0$ is excluded from these leading coefficient statements.

For F, every fixed $b > 0$ and $2 < h < 4$ give a different explicit window:

$$D_M^F(\ell - b \log \ell) = 2 C_{\nu,\sigma} \ell^{-2b} [1 + O_h(\ell^{-(h-2)b})]. \tag{17.15}$$

These formulas follow by inserting the Section 13–15 tails in the mirror identity and choosing q in (17.9) larger than the power being resolved. For (17.11), the fast tail relative to the slow one is $O(\ell^{-n(1-\alpha)})$; for (17.15) it is exponentially small in ℓ. No unknown δ_M appears in the time choices.

For a fixed gap $g > 0$, (16.13) transfers to the exact D_M^g with its displayed $O(s^{-5} + k^{-5})$ error whenever s, k are fixed positive powers of ℓ or fixed fractions away from zero. This is an absolute envelope result. A relative statement still requires separation from zeros of the relevant oscillatory combination, and the estimate is not uniform as $g \downarrow 0$.

The asymptotic window shapes are explicit, but these theorems do not give a numerical mass at which a chosen percentage accuracy begins. They are restricted to the stated free-matter, $G_N \to 0$ model and do not identify a finite-gravity chaos exponent. In particular the M^{-4} mirror correction for F at fixed T need not be the leading exact correction; the above error bound need not resolve that power.

17.5 Diagnostic scope and next physical comparison

The new diagnostics check the shifted Gamma ratios on growing-parameter fixtures, the explicit Euler derivative inequality and exact-energy coordinate estimates. These probe the ingredients and conventions of the proof. They do not certify its supremum constants, perform a full finite-M 6j packet integration, or independently validate the proof. Previously frozen limiting spectra are not recomputed.

The saved run contains 290 checks at 80 decimal digits, with 110-digit Gamma-ratio cross-checks. On the recorded shifts up to $Q_0 = 64$, the largest $M|E|/(1 + Q_0)^2$ is about 0.080004; this is a sampled quantity, not the universal constant in (17.3). The largest sampled ratio of a Gauss derivative to its Euler-integral bound is about 0.883322. The energy-coordinate and normalized tail inequalities are checked separately. No finite collection of these values proves the growing-box supremum estimate.

With the explicit asymptotic window established at this level, the next physical step is a matched finite-G_N comparison. The external state, projection, proper time, normalization, matter representation and contour must agree before the current preparation effects can be compared with a genuine gravitational interaction or a microscopic candidate. Numerically useful error constants and exact finite-mass packet checks remain a separate validation task.

18 Open questions

The weak realization fixes the space and the order of regulation. Section 6 removes auxiliary position and group cutoffs for the creation pairing on explicit test seeds and arbitrary physical inputs. Section 7 identifies annihilation as an integrated mode limit on all physical inputs and as an absolute uncut position formula on seed inputs. Pointwise convergence and absolute position formulas for arbitrary physical inputs remain open. The Gaussian proper-time smearing and observer threshold have not been removed.

Two further questions are central to the original holographic objective. First, Sections 12 and 13 give the logarithmic-time limit and distinguish the fixed-protocol exponential early correction from an inverse-cubic hard-boundary correction for an exponential packet. Section 16 retains both oscillatory branches and proves uniform absolute approximation on the full time axis. Its mirror identity determines the fast-branch tail. Section 17 supplies their exponential-polynomial growth bounds and explicit asymptotic windows for packets with a positive exponential moment. A numerically certified onset mass and full exact finite-mass packet checks remain open. Section 15 classifies the stated integer-order Gamma densities and gapped exponential packets. General densities, degenerate real coefficients, and a uniform gap-closing crossover still require separate analysis. Section 14 supplies a numerically resolved

moving-protocol curve for the exponential packet, with empirical quadrature diagnostics rather than certified total error bars. The constant source time-origin discrepancy is now isolated algebraically, but is not an author-confirmed correction. Second, a finite-G_N or microscopic comparison must match states, spectral weights, time conventions, external legs, and integration contours in a common parameter regime. The results here do not settle either question. The known noncrossed large-mass expansion and the finite-G_N comparison notes remain separate parts of the research project.

A Observer wave packets and spectral normalization

This appendix supplies the positive/negative frequency normalization without formally multiplying generalized eigenfunctions and delta functions. Fix $k = |n|$, $h = k + 1/2$, and $r(s) = s\tanh \pi s$. In the phase convention of [1, (3.11), (4.9), (A.5)–(A.6)],

$$v_s(t) := \sqrt{\cosh t}\, \Psi_n^s(t,0) = N_k(s)\cosh^h t\, \mathbf{F}(h-is, h+is; k+1; z(t)), \tag{A.1}$$

$$z(t) = \frac{1 - i\sinh t}{2}, \tag{A.2}$$

$$N_k(s) = \sqrt{\frac{r(s)}{\pi}}\, e^{i\phi_{-k,s}/2} \frac{(1-i)(2i)^{-k}\Gamma(1/2-is)}{2\sqrt{2\cosh \pi s}\,\Gamma(1/2-k-is)}. \tag{A.3}$$

Here $\mathbf{F} = {}_2F_1/\Gamma(c)$ and the half phase is chosen smoothly and compatibly with the $n \leftrightarrow -n$ relation. Gamma recurrence gives

$$|N_k(s)|^2 = \frac{r(s)2^{-2k}}{4\pi^2}|\Gamma(h+is)|^2. \tag{A.4}$$

For $t = \epsilon\tau$, $\epsilon = \pm 1$, $\tau > 0$, set $x = e^{-\tau}$. The identity $-z(\epsilon\tau) = i\epsilon(1+i\epsilon x)^2/(4x)$ and the hypergeometric connection formula [6, 15.8.2] give

$$v_s(\epsilon\tau) = \sum_{\sigma=\pm 1} A_{\epsilon\sigma}(s) e^{i\sigma s\tau} M_{\epsilon\sigma}(x,s), \tag{A.5}$$

$$A_{\epsilon\sigma}(s) = B_\sigma(s) e^{-i\epsilon\pi h/2} e^{-\epsilon\sigma\pi s/2}, \tag{A.6}$$

$$B_\sigma(s) = N_k(s) 2^{h-2i\sigma s} \frac{\Gamma(2i\sigma s)}{\Gamma(h+i\sigma s)\Gamma(1/2+i\sigma s)}, \tag{A.7}$$

$$M_{\epsilon\sigma}(x,s) = (1+x^2)^h (1+i\epsilon x)^{-2q_\sigma}\, {}_2F_1\left(q_\sigma, \tfrac{1}{2} - i\sigma s; 1-2i\sigma s; \frac{4i\epsilon x}{(1+i\epsilon x)^2}\right), \tag{A.8}$$

$$q_\sigma = h - i\sigma s. \tag{A.9}$$

The linear terms in the last expression cancel, so $M = 1 + x^2 Q$ with Q analytic near $x = 0$ on every compact positive spectral interval. For fixed spectral and time derivative orders, $|\partial_s^j \partial_\tau^l (M-1)| \le C e^{-2\tau}$ for $\tau \ge \log 32$. Including the phase in the differentiated remainder adds a factor $(1+\tau)^j$. Thus $\int f(s) v_s(t)\, \mathrm{d}s$ is Schwartz for $f \in C_c^\infty((0,\infty))$.

The standard gamma moduli give

$$|\Gamma(2is)|^2 = \frac{\pi}{2s\sinh 2\pi s}, \qquad |\Gamma(1/2+is)|^2 = \frac{\pi}{\cosh \pi s}, \qquad |B_\sigma(s)|^2 = \frac{1}{8\pi^2 \cosh \pi s}. \tag{A.10}$$

Now use the unitary radial variable $r_s(t) = \sqrt{2\pi}\, v_s(t)$ and write $W_f = \int r_s f(s)\, \mathrm{d}s$. Its Schwartz property and differential equation imply

$$e^{-iTH_k} W_f = W_{e^{-iTs^2} f}. \tag{A.11}$$

As $T \to +\infty$, on each half-axis only the $\sigma = +$ outgoing free packet remains. The $\sigma = -$ phase has no stationary point there, and repeated spectral integration by parts bounds its L^2 norm by a quantity tending to zero. The exponentially small Jost remainder tends to zero on compact positions by nonstationary phase, and its tail is uniformly L^2 dominated. The outgoing packet restricted to that half-axis has the full-line Fourier norm in the limit; its omitted half-line is again nonstationary. Plancherel therefore gives the packet Gram density

$$4\pi^2 \sum_{\epsilon=\pm 1} \overline{A_{\epsilon,+}(s)} A'_{\epsilon,+}(s). \tag{A.12}$$

For two positive columns, (A.10) makes this density one. For the negative-frequency column, $A'_{\epsilon,+} = (-1)^n A_{-\epsilon,+}$. Its cross density is

$$8\pi^2 (-1)^n |B_+(s)|^2 \cos(\pi h) = 0. \tag{A.13}$$

The negative column has the same unit norm. Different n are already orthogonal in θ.

Finally the two asymptotic coefficient columns at $+\infty$ have determinant

$$-2(-1)^n B_+ B_- \sinh \pi s \neq 0.$$

They span the two-dimensional continuous solution space and are a smooth invertible change of basis from the scattering modes on every compact positive interval. The bound modes are orthogonal to these packets by the self-adjoint radial spectral decomposition. This proves the full continuous normalization and completeness needed in Section 4; it makes no assertion that the entire position representation lacks discrete series.

B Local vertex conventions and the crossed prefactor

We distinguish the vertex reconstructed from the mode integral from the literal expression in (A.7) of [1]. This appendix fixes the convention used in (5.1); it is not an author-confirmed erratum. Set $r_j = s_j \tanh \pi s_j$, $c_j = \cosh \pi s_j$, and define

$$\begin{aligned} G_4(s_1, s_2, s_3) &= \Gamma\Big(\tfrac{1}{4} + \tfrac{i}{2}(-s_1 + s_2 + s_3)\Big)\, \Gamma\Big(\tfrac{1}{4} + \tfrac{i}{2}(-s_1 - s_2 + s_3)\Big) \\ &\quad \times \Gamma\Big(\tfrac{1}{4} + \tfrac{i}{2}(-s_1 + s_2 - s_3)\Big)\, \Gamma\Big(\tfrac{1}{4} + \tfrac{i}{2}(s_1 + s_2 + s_3)\Big), \end{aligned} \tag{B.1}$$

$$D(s_1, s_2, s_3) = \Gamma(1/2 - i s_1)\Gamma(1/2 + i s_2)\Gamma(1/2 + i s_3). \tag{B.2}$$

Then the reconstructed vertex is

$$\mathcal{V}^{s_1}_{s_2,s_3,\mathrm{rec}} = e^{-i\pi/4} \frac{c_1 \sqrt{r_1 r_2 r_3}}{8\pi \sqrt{c_1 c_2 c_3}} \frac{G_4}{D} = \frac{\mathcal{V}^{s_1}_{s_2,s_3,\mathrm{literal}}}{\Gamma(1/2 + i s_3)^2}. \tag{B.3}$$

The second equality states the discrepancy explicitly.

Here is its analytic normalization check. Write

$$P_j(z) = P_{-1/2+is_j}(z), \qquad \Psi_0^{s_j}(t) = \frac{\sqrt{r_j}\, P_j(i \sinh t)}{2\sqrt{\pi c_j}}.$$

Closing the vertical contour to the right and using the Legendre jump relation gives

$$\int_{\mathbb{R}} \cosh t\, P_1(-i \sinh t) P_2(i \sinh t) P_3(i \sinh t)\, \mathrm{d}t = 2c_1 \int_1^\infty P_1(u) P_2(u) P_3(u)\, \mathrm{d}u. \tag{B.4}$$

The endpoint circle is $O(\epsilon|\log\epsilon|)$ and the large semicircle is $O(R^{-1/2})$, for fixed positive labels. The triple-Legendre identity of [10] is

$$\int_1^\infty P_1P_2P_3\,\mathrm{d}u=\frac{\Gamma_8\,c_1c_2c_3}{4\pi^5},\qquad \Gamma_8=\prod_{\epsilon_j=\pm1}\Gamma\left(\tfrac14+\tfrac{i}{2}\sum_{j=1}^{3}\epsilon_j s_j\right). \tag{B.5}$$

Including the angular integral yields

$$T_{000}=\frac{c_1\sqrt{r_1r_2r_3}\sqrt{c_1c_2c_3}}{8\pi^{11/2}}\Gamma_8. \tag{B.6}$$

The circle three-point integral in the same convention gives $C^{\mathrm{phys}}_{S,000}=e^{-i\pi/4}G_4/(\pi^{3/2}D)$. Dividing by its conjugate proves (B.3) at zero magnetic labels. It fixes all finite-mode vertices as follows: the complete principal three-leg invariant has S and A channels; the position integral is even under simultaneous angular reflection, while the explicit A-channel sign product is odd. The A coefficient vanishes and the nonzero $S,000$ coefficient fixes the remaining scalar. This uses the stated reflection and half-phase conventions; the wording of footnote 33 in [1] should not be used to override its explicit antisymmetric kernel. Continuity on compact positive label sets extends the almost-everywhere decomposition identity to fixed positive labels.

For the crossing graph put $(s_1,s_2,s_3,s_4)=(d,a,b,c)$ and let $A=\nu_A$, $B=\nu_B$. The four scalar factors in (B.2) of [1] are

$$P_{\mathcal{V}}=\overline{\mathcal{V}^{d}_{A,a,\mathrm{rec}}}\,\overline{\mathcal{V}^{a}_{B,b,\mathrm{rec}}}\,\mathcal{V}^{c}_{A,b,\mathrm{rec}}\,\mathcal{V}^{d}_{B,c,\mathrm{rec}}. \tag{B.7}$$

Their half phases cancel. Define the sixteen-gamma numerator explicitly by

$$\Gamma^B_{16}=\overline{G_4(d,A,a)}\,\overline{G_4(a,B,b)}G_4(c,A,b)G_4(d,B,c). \tag{B.8}$$

The six pairs of denominator gamma functions give $\prod_{s=d,a,b,c,A,B}c_s/\pi^6$. Thus

$$P_{\mathcal{V}}=\frac{r_dr_ar_br_cr_Ar_B\,c_d^2c_ac_c}{4096\pi^{10}}\Gamma^B_{16}. \tag{B.9}$$

The numerator is exactly the collection of arguments in (B.27).

Let $S(d,a,b,c,A,B)$ denote the raw recoupling function in (B.17), with its stated conjugation orientation. In normalized trees the S/S subkernel acquires $\sqrt{w(a)w(c)}$. It is a subkernel of the complete unitary Racah transform, not by itself a unitary transform. The scalar local factors contribute the inverse $w(d)\sqrt{w(a)w(c)}$, together with $\pi^2/(r_Ar_B)$. Using the physical measure $\rho_{\mathrm{Pl}}(d)$ and the Hartle–Hawking weight $\mathrm{sech}^2(\pi d)$, (B.9) gives

$$\frac{r_dr_ar_br_c\,c_ac_c}{4096\pi^{11}}\Gamma^B_{16}S(d,a,b,c,A,B). \tag{B.10}$$

This is (B.25)–(B.27), with a,c carrying sinh and b,d carrying tanh. The calculation uses normalized tree inner products, not an uncancelled $\delta(0)$ or group volume.

Conjugation orientation can be checked independently. The original modes are fixed by the antiunitary map $f(t,\theta)\mapsto\overline{f(-t,-\theta)}$. Applying it to the absolutely convergent integral (5.1) shows that each zero-time finite-mode local matrix element is real. The complete zero-time contracted spectral kernel is therefore real as well, by normalized spectral limits. Reversing bra and ket conjugates both the four vertex factors and the Racah kernel and gives the same zero-time result. It does not justify conjugating a nonzero-time phase independently.

For equal matter labels, the assembled symmetric function $\mathcal{S}^\nu$ in (A.14) can be cyclically relabeled to (a,b,c,d). The raw S or its gamma factor alone cannot be relabeled this way. This distinction is necessary when comparing the local construction with the archived implementation of the crossed kernel.

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